

Logs

Day 1

Date	Day	Task	Task Description	Work Done	Hours
25-Apr-23	Tuesday	Orientation	1. Recorded the company's culture fostering collaboration, diversity, and innovation in the timesheet entry. 2. Noted the goals of delivering outstanding client solutions and pursuing a 20% reduction in environmental impact over the upcoming three years in the timesheet entry.	Company overview presentation	2
		Team Introduction	1. Conducted a team meeting to facilitate introductions and enhance familiarity among team members. 2. During the meeting, each team member briefly introduced themselves, sharing their roles and a bit about their background and expertise.	Team members' roles and backgrounds	2
		Platform Access Setup	1. Established access for team members to essential platforms, ensuring they had the necessary tools for their tasks. 2. Assisted team members in navigating platform setup, granting permissions, and resolving any access-related issues to streamline their workflow.	Account creation and access setup	2
		Familiarization	1. Engaged in an exploration of the resources and tools provided, aiming to familiarize the team with their functionalities and potential applications. 2. Collaboratively assessed the capabilities of the resources and tools, identifying how they could be effectively integrated into our projects and tasks.	Browsing through available tools	2

Day 2

Date	Day	Task	Task Description	Work Done	Hours
26-04-2023	Wednesday	Platform Orientation	1. Received comprehensive platform orientation, gaining a thorough understanding of their interfaces, features, and navigation. 2. The orientation sessions equipped me with the knowledge to utilize these platforms adeptly for streamlined project management and seamless collaboration.	Platform overview and navigation	2

		Accessing Data	1. Facilitated by team members the process of learning how to access project datasets, ensuring team members had the necessary knowledge to retrieve crucial data. 2. Assisted by team members in navigating the dataset repositories, sharing step-by-step instructions to retrieve specific datasets required for their tasks.	Navigating data repositories	2
		Environment Setup	1. Successfully configured the development environment with assistance from team members, ensuring all required software and settings were properly established. 2. Collaboratively worked with team members to install and configure relevant tools, libraries, and dependencies, resulting in a productive and effective development environment.	Setting up required tools	2
		Collaborative Tools Setup	1. Established communication and collaboration channels, enabling seamless interactions among team members. 2. Implemented tools and platforms that allowed team members to easily share updates, exchange ideas, and collaborate on projects, fostering a cohesive and efficient working environment.	Creating accounts, configuring	2

Day 3

Date	Day	Task	Task Description	Work Done	Hours
27-04-2023	Thursday	Project Scope Overview	1. Presented a comprehensive breakdown of the project's overarching objectives, elucidating the desired outcomes and specific targets to be achieved. 2. Highlighted the key milestones and deliverables that align with the project's objectives, offering a clear roadmap for the team's efforts. 3. Emphasized the significance of each objective in relation to the project's overall mission, ensuring team members had a thorough understanding of their roles in achieving these goals.	Discussion on project scope	2
		Task Breakdown	1. Deconstructed the project into distinct tasks, outlining each step required to accomplish the overall objectives. 2. Established clear milestones within the project timeline, serving as checkpoints to gauge progress and ensure alignment with project goals.	Identifying key project components	2

			3. Collaborated with team members to assign responsibilities, set deadlines, and create a structured plan that facilitates efficient task completion and keeps the project on track.		
		Expectation Alignment	1. Engaged in a thorough discussion to outline the specific goals of the internship, encompassing both the skills to be acquired by the intern and the contributions they would make to the team. 2. Defined measurable outcomes for the internship period, ensuring a clear understanding of the expectations and providing a framework for evaluating the intern's progress and achievements	Discussion on internship goals	2
		Task Assignment	1. Collaboratively identified the initial task owners, ensuring a balanced distribution of responsibilities based on individual strengths and expertise. 2. Allocated tasks to team members based on their skill sets and availability, ensuring an efficient and focused start to the project. 3. Facilitated a transparent discussion among team members to establish clear ownership of tasks, enhancing accountability and minimizing potential overlaps or gaps in responsibilities.	Identifying initial task owners	2

Day 4

Date	Day	Task	Task Description	Work Done	Hours
28-04-2023	Friday	Software Installation	Successfully installed the essential software and tools required for the project, guaranteeing a smooth workflow and enabling team members to begin their tasks. Ensured that all team members had access to the necessary software and tools by providing step-by-step instructions for installation and offering support in case of any technical challenges.	Installing Python, IDE, etc.	2
		Platform Integration	Collaboratively integrated various platforms to streamline data access, allowing team members to retrieve and utilize project datasets seamlessly. Ensured a smooth synchronization between platforms, facilitating efficient data sharing and reducing the time required for accessing critical information. Verified the compatibility and functionality of integrated platforms, addressing any potential	Connecting to cloud platforms	2

			issues and ensuring that data access was consistently reliable for all team members.C10		
		Data Exploration	Conducted an initial exploration of the provided datasets to gain insights into their structure, contents, and potential relevance to the project. Analyzed the datasets to identify any patterns, anomalies, or initial trends that could guide the project's direction and decision-making. Collaborated with team members to share findings and observations from the dataset exploration, fostering a collective understanding and setting the stage for further analysis.	Browsing data repositories	2
		Coding Environment	Successfully set up the coding environment for the project, configuring necessary IDEs, libraries, and dependencies to facilitate efficient development. Ensured that team members had a consistent and well-prepared coding environment by providing guidance on installation steps and troubleshooting common issues.	Configuring coding environment	2

Day 4

Date	Day	Task	Task Description	Work Done	Hours
28-04-2023	Friday	Software Installation	Successfully installed the essential software and tools required for the project, guaranteeing a smooth workflow and enabling team members to begin their tasks. Ensured that all team members had access to the necessary software and tools by providing step-by-step instructions for installation and offering support in case of any technical challenges.	Installing Python, IDE, etc.	2

		Platform Integration	Collaboratively integrated various platforms to streamline data access, allowing team members to retrieve and utilize project datasets seamlessly. Ensured a smooth synchronization between platforms, facilitating efficient data sharing and reducing the time required for accessing critical information. Verified the compatibility and functionality of integrated platforms, addressing any potential issues and ensuring that data access was consistently reliable for all team members.C10	Connecting to cloud platforms	2
		Data Exploration	Conducted an initial exploration of the provided datasets to gain insights into their structure, contents, and potential relevance to the project. Analyzed the datasets to identify any patterns, anomalies, or initial trends that could guide the project's direction and decision-making. Collaborated with team members to share findings and observations from the dataset exploration, fostering a collective understanding and setting the stage for further analysis.	Browsing data repositories	2
		Coding Environment	Successfully set up the coding environment for the project, configuring necessary IDEs, libraries, and dependencies to facilitate efficient development. Ensured that team members had a consistent and well-prepared coding environment by providing guidance on installation steps and troubleshooting common issues.	Configuring coding environment	2

Week 1

Date	Day	Task	Task Description	Work Done	Hours
01-05-2023	Monday	Project Kick-off	1. Provided a comprehensive introduction to the project's overarching goals and scope, outlining the intended outcomes and the specific boundaries of the project. 2. Presented a clear outline of the project's objectives, ensuring that all team members understood the purpose and direction of their efforts. 3. Discussed the scope limitations to	Review project objectives and plan	2

			avoid potential scope creep, ensuring that team members were aligned on the project's boundaries and the tasks that fell within its defined scope.		
		Environment Setup	1. Created a virtual environment for the project, isolating dependencies and ensuring a controlled development environment. 2. Installed required packages within the virtual environment, enabling team members to work with specific libraries and tools without conflicts. 3. Provided step-by-step instructions for creating the virtual environment and installing necessary packages, ensuring consistency and ease of setup across the team.	Set up Python environment and install packages and their dependencies	1
		Data Collection	1. Retrieved raw e-commerce datasets from BigQuery, sourcing the foundational data needed for analysis and further processing. 2. Executed SQL queries to extract relevant data from BigQuery, ensuring that the acquired datasets encompassed the necessary information for the project's objectives.	Download customer, product and events data	1
		Data Loading and Inspection	1. Utilized Pandas to load the acquired datasets, converting them into a structured format within Python for easy manipulation and analysis. 2. Conducted an initial examination of the loaded data, performing data profiling to gain insights into its structure, quality, and characteristics. 3. Employed Pandas' functionalities to perform basic exploratory data analysis (EDA), such as summarizing statistics, identifying missing values, and visualizing data distributions. 4. Collaborated with team members to share observations and insights from the initial data examination, paving the way for informed decision-making and further analysis.	Import data, view initial stats	2
02-05-2023	Tuesday	Data Preprocessing	1. Identified missing values within the dataset and implemented strategies such as imputation or removal to address these gaps and ensure data completeness. 2. Detected outliers in the data using statistical methods or visualization techniques, evaluating their impact on the analysis and deciding whether to keep, transform, or exclude them. 3. Collaborated with team members to	Clean data, handle nulls	4

			make informed decisions about handling missing values and outliers, ensuring consistency and accuracy across the analysis. 4. Applied appropriate techniques to impute missing values, considering factors such as data type, distribution, and domain knowledge. Documented the steps taken to handle missing values and outliers, providing transparency and reproducibility for the analysis process.		
		Exploratory Data Analysis	1. Created a variety of data visualizations, including bar charts, scatter plots, and histograms, to visually represent the dataset's patterns and relationships. 2. Utilized libraries like Matplotlib or Seaborn to generate the visualizations, enhancing the team's understanding of the data's distribution and trends. 3. Collaborated with team members to choose the most effective visualization techniques for conveying insights and supporting the project's objectives.	Create histograms, scatter plots	2
		Feature Identification	1. Explored the dataset to identify potential features that could contribute to the analysis and align with the project's goals. 2. Conducted preliminary feature selection, considering factors such as relevance, uniqueness, and potential impact on the desired outcomes. 3. Collaborated with domain experts and team members to validate the selected features and ensure their alignment with the analysis objectives. 4. Documented the rationale behind the chosen features, providing a clear record of the decision-making process for future reference.	Select relevant customer attributes	2

03-05-2023, 04-05-2023	Wednesday , Thursday	Initial Model Exploration	1. Conducted thorough research to explore a range of Machine Learning (ML) algorithms suitable for collaborative filtering, considering techniques such as matrix factorization, user-item embeddings, and neighborhood-based methods. 2. Investigated advanced collaborative filtering algorithms like Singular Value Decomposition (SVD), Alternating Least Squares (ALS), and factorization machines, evaluating their strengths and weaknesses for the project's context. 3. Explored hierarchical filtering methods, delving into decision tree-based algorithms like Random Forests and Gradient Boosting, which offer the potential to capture intricate relationships in data with hierarchical structures. 4. Analyzed the applicability of deep learning models such as Recurrent Neural Networks (RNNs) or Transformers for hierarchical filtering tasks, considering their ability to capture sequential dependencies. 5. Collaborated with team members to discuss the feasibility and suitability of various ML algorithms, taking into account factors such as data size, complexity, and interpretability. 6. Considered the scalability of algorithms, particularly for large datasets, and their potential to be integrated into the existing infrastructure. 7. Evaluated algorithm performance metrics such as accuracy, precision, recall, and F1-score to determine the effectiveness of different models in achieving the desired outcomes. 8. Reviewed research papers, articles, and online resources to stay updated on the latest advancements in collaborative filtering and hierarchical filtering techniques. 9. Documented the findings of the research, providing a comprehensive overview of the considered ML algorithms, their key characteristics, and potential use cases. 10. Presented the research outcomes to the team, fostering informed discussions and decisions on the most suitable ML algorithms for	Investigate collaborative filtering	20
---------------------------	-------------------------	------------------------------	---	-------------------------------------	----

			implementing collaborative and hierarchical filtering in the project.		
05-05-2023	Friday	Task Review and Planning	1. Conducted a comprehensive review of the tasks accomplished during Week 1 and their outcomes, identifying any challenges or adjustments required for the upcoming week. 2. Collaboratively planned tasks for Week 2, considering the project's timeline and priorities, and allocated responsibilities among team members to ensure efficient progress.	Prepare for next week's activities	1
		Model Selection Research	1. Conducted thorough research to identify potential machine learning models suitable for addressing the project's objectives and challenges. 2. Explored a range of models including regression, classification, clustering, and neural networks, evaluating their applicability to the specific dataset and problem domain. 3. Considered both traditional statistical models and modern deep learning architectures, weighing factors such as model complexity, interpretability, and computational requirements. 4. Collaborated with domain experts and team members to validate the suitability of identified models based on their expertise and insights. 5. Documented the research findings, including the characteristics, advantages, and limitations of each model, to guide the decision-making process and ensure transparency in model selection.	Investigate content-based models	4
		Communication and Updates	1. Conducted a team meeting to provide updates on the progress made, sharing insights on accomplished tasks, challenges faced, and solutions implemented.	Discuss findings, share insights	1

			2. Collaborated with team members to ensure everyone was aligned on the project's current status and discussed potential adjustments to the plan based on the progress achieved.		
--	--	--	--	--	--

Week 2

Date	Day	Task	Task Description	Work Done	Hours
08-05-2023	Monday	Feature Extraction	1. Analyzed the available data to identify relevant features and attributes that could provide insights into customer behavior, such as purchase history, browsing patterns, and interaction frequencies. 2. Utilized data transformation techniques to extract customer behavior information from raw data, aggregating and structuring it in a way that facilitates analysis. 3. Explored techniques like cohort analysis, sessionization, and user segmentation to gain a deeper understanding of customer behavior patterns and trends. 4. Collaborated with data engineers to ensure the efficient extraction of required data, addressing any technical challenges that arose during the process. 5. Documented the extraction process, outlining the methods used, data sources accessed, and any preprocessing steps applied to ensure transparency and reproducibility.	Extract key attributes	4
		Data Transformation	1. Conducted scaling of numerical features to establish uniformity for effective variable comparison. 2. Observed heightened model performance, particularly with algorithms sensitive to varying scales, after implementing scaling techniques. 3. Safeguarded inherent data patterns during scaling procedures, upholding data integrity and minimizing potential distortions. 4. Adhered to algorithm specifications by selecting appropriate scaling methods like normalization or standardization, fostering optimal model interaction. 5. Ensured seamless integration of categorical variables into the model by performing essential encoding (one-hot/label), enabling accurate representation in numerical form.	Prepare data for modeling	4

09-05-2023, 10-05-2023	Tuesday, Wednesday	Model Selection	1. Investigated diverse recommendation algorithms to identify optimal choices for specific use cases and data characteristics. 2. Analyzed algorithmic strengths and weaknesses, considering factors like scalability, personalization, and data sparsity. 3. Explored collaborative filtering, content-based, and hybrid approaches to determine the most suitable strategies for delivering accurate and effective recommendations.	Investigate collaborative filtering	2
		Model Implementation	1. Imported necessary libraries such as NumPy, Pandas, and Scikit-learn for data manipulation and computations. 2. Loaded the dataset and preprocessed it. This involved handling missing values, transforming data into a user-item interaction matrix, and splitting data into training and testing sets. 3. Chose a similarity metric coefficient to measure the similarity between users or items. 4. Calculated the similarity matrix based on the chosen metric. This involved computing similarities between users or items using the interaction matrix. 5. For user-based collaborative filtering, predicted ratings or preferences for items that a user hadn't interacted with by using weighted averages of similar users' ratings. 6. For item-based collaborative filtering, predicted ratings for items using similar items' ratings and user-item interactions. 7. Evaluated the performance of the collaborative filtering model using appropriate evaluation metrics like Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE). 8. Experimented with hyperparameters such as neighborhood size (for user-based CF) or the number of similar items (for item-based CF) to optimize the model's performance. 9. Implemented solutions to handle the "cold start" problem for new users or items that lacked sufficient interaction data.	Develop an initial recommendation model	16
11-05-2023	Thursday	Model Evaluation	1. Calculated accuracy to assess the overall correctness of predictions in relation to actual outcomes. 2. Computed precision, evaluating the	Analyze model metrics	2

			proportion of correctly predicted positive instances among all predicted positives. 3. Evaluated recall, indicating the fraction of actual positive instances that were correctly predicted among all actual positives.		
		Model Tuning	1. Explored various hyperparameters to fine-tune model performance. 2. Utilized techniques like grid or random search for optimization. 3. Employed cross-validation to choose optimal hyperparameter values.	Tune the model for better results	2
		Cross-Validation	1. Employed cross-validation to assess the model's performance reliably across different subsets of the data. 2. Split the dataset into multiple folds, training the model on some folds and testing on others iteratively. 3. Calculated evaluation metrics such as accuracy, precision, recall, or F1-score for each fold to gauge consistency. 4. Derived an aggregated performance estimate, offering a comprehensive view of the model's effectiveness.	Assess model performance	2
12-05-2023	Friday	Model Updates	1. Reviewed monitoring data to gain insights into system behavior and performance. 2. Identified patterns, trends, and anomalies within the collected data. 3. Proposed updates based on the analysis to improve system efficiency, reliability, or security. 4. Ensured that the suggested updates align with the identified issues and contribute to overall enhancement.	Identify areas for enhancement	2
		Performance Optimization	1. Incorporated recommended modifications to the system based on identified areas for improvement. 2. Executed changes to enhance functionality, performance, or user experience. Monitored the effects of implemented enhancements to validate their impact and ensure better results.	Optimize model performance	2
		Evaluation of the Model	1. Assessed the model's performance following hyperparameter optimization to gauge its improvements. 2. Utilized relevant evaluation metrics, such as accuracy, precision, recall, or F1-score, to measure the enhanced model's effectiveness. 3. Compared the post-optimization results with the baseline to validate the significance of the improvements achieved.	Update optimization and results	2

		Task Review and Planning	1. Conducted a comprehensive review of tasks completed in Week 2 to track progress and identify any pending items. 2. Developed a task plan for Week 3, outlining objectives, priorities, and timelines to ensure efficient and organized workflow.	Prepare for next week's activities	1
		Communication and Updates	1. Scheduled a team meeting to discuss ongoing projects and provide status updates. 2. Shared progress achieved, challenges encountered, and next steps to ensure team alignment and effective collaboration.	Discuss findings, share insights	1

Week 3

Date	Day	Task	Task Description	Work Done	Hours
15-05-2023	Monday	Model Hyperparameter Tuning	1. Identified key hyperparameters within the model that significantly influence its behavior and outcomes. 2. Utilized techniques such as grid search or random search to explore a range of hyperparameter values. 3. Evaluated model performance using metrics like cross-validation accuracy or loss across different hyperparameter configurations. 4. Selected the best-performing hyperparameters to enhance the model's overall performance and generalization capabilities.	Tune the model for better results	2
		Bug Fixes and Improvements	1. Reviewed the model's codebase to identify and address any errors or bugs affecting its functionality. 2. Analyzed error logs, exception messages, and debugging tools to pinpoint the root causes of issues. 3. Refined data preprocessing steps to ensure high-quality input for the model. 4. Conducted exploratory data analysis to gain insights into potential data patterns and anomalies. 5. Experimented with different algorithms or architectures to identify the best-fit solution for the given problem. 6. Adjusted hyperparameters to fine-tune the model's behavior and	Address issues, improve efficiency	12

			optimize performance. 7. Applied techniques such as regularization or dropout to mitigate overfitting and improve generalization. 8. Implemented feature engineering strategies to extract meaningful information from the data. 9. Validated changes using cross-validation or separate test datasets to quantify improvements. Documented all enhancements and changes made, ensuring a clear record for future reference.		
16-05-2023	Tuesday	Cross-Validation	1. Partitioned the dataset into multiple subsets using cross-validation. 2. Trained the model on a subset while validating it on the remaining subsets iteratively. 3. Calculated performance metrics to assess the model's consistency and generalization ability across different data splits.	Assess model performance	2
		Model Evaluation	1. Calculated accuracy to assess the overall correctness of predictions in relation to actual outcomes. 2. Computed precision, evaluating the proportion of correctly predicted positive instances among all predicted positives. 3. Evaluated recall, indicating the fraction of actual positive instances that were correctly predicted among all actual positives.	Analyse model metrics	2
		Model Updates	1. Reviewed collected monitoring data to gain insights into system performance and behavior. 2. Identified trends, anomalies, and potential areas for improvement within the data. 3. Formulated recommendations for updates or adjustments to enhance system efficiency, reliability, or security based on the analysis.	Identify areas for enhancement	2
17-05-2023	Wednesday	Performance Optimization	1. Incorporated the suggested improvements or updates based on the analysis. 2. Modified the relevant components of the system to align with the enhancement plan. 3. Ensured seamless integration of the enhancements within the existing system architecture. 4. Conducted thorough testing to verify that the implemented changes produce the desired	Optimize model performance	8

			outcomes. 5. Monitored the system's performance after implementation to validate the effectiveness of the enhancements. 6. Gathered feedback from users or stakeholders to assess their satisfaction and gather insights for further refinement.		
18-05-2023, 19-05-2023	Thursday, Friday	Task Review and Planning	1. Conducted a comprehensive review of completed tasks and progress made during Week 3. 2. Analyzed any pending items and identified areas needing further attention. 3. Developed a task plan for Week 4, outlining objectives, priorities, and timelines to guide efficient work allocation and successful project advancement.	Prepare for next week's activities	2
		Communication and Updates	1. Scheduled a team meeting to provide updates on project progress and developments. 2. Shared information about achieved milestones, challenges faced, and upcoming tasks to maintain team cohesion and ensure everyone is aligned.	Discuss findings, share insights	1
		Model Deployment Preparation	1. Finalized the trained model by saving its parameters and architecture for later use. 2. Packaged the necessary code, libraries, and dependencies to ensure the model's functionality in a deployment environment. 3. Tested the model in the deployment environment to verify its compatibility and responsiveness before making it accessible to users.	Package model for deployment	2
		Model Deployment	1. Utilized Google Cloud's Vertex AI platform to facilitate model deployment. 2. Packaged the trained model along with any required preprocessing and post-processing scripts. 3. Configured model deployment settings, including hardware specifications and scaling options. 4. Monitored the deployed model's performance and usage, making necessary adjustments to ensure optimal functionality.	Set up model deployment, deploy	2
		Real-Time Inference Testing	1. Sent sample data inputs to the deployed model to receive real-time predictions. 2. Verified the accuracy and timeliness of the model's	Interact with deployed model	2

			predictions to ensure its reliability in a real-world scenario.		
		Bug Fixes and Enhancements	1. Analyzed deployment logs and performance metrics to identify potential bottlenecks or inefficiencies. 2. Adjusted deployment settings such as resource allocation and scaling based on observed usage patterns. 3. Employed load testing to simulate varying levels of user activity and evaluate how the deployed model handles increased demand. 4. Continuously monitored the deployed model's performance and response times, iteratively refining deployment configurations for optimal efficiency.	Address any issues, and enhance	3

Week 4

Day	Day	Task	Task Description	Work Done	Hours
22-05-2023	Monday	Model Export and Packaging	1. Finalized model training and ensured it meets desired performance criteria. Saved the trained model's weights, architecture, and necessary preprocessing steps. 2. Created APIs or functions to facilitate communication between the deployed model and external applications. 3. Packaged model files, dependencies, and required environment configurations for deployment. 4. Conducted thorough testing to confirm the model's functionality, stability, and compatibility in the deployment environment.	Package model for deployment	3
		Model Deployment	1. Utilized Google Cloud's Vertex AI platform to deploy the model. 2. Configured deployment settings such as instance type, autoscaling, and authentication for secure and efficient model hosting.	Set up model deployment, deploy	2
		Real-Time Inference Testing	1. Generated sample data inputs representative of real-world scenarios. 2. Sent these sample inputs to the deployed model's API to obtain real-time predictions. 3. Examined prediction outputs for correctness and consistency with	Interact with deployed model	4

			expected results. 4. Assessed prediction response times to ensure prompt and efficient service. Iteratively refined and retested as needed to validate the reliability of real-time predictions.		
23-05-2023	Tuesday	Bug Fixes and Enhancements	1. Monitored deployed model's performance, tracking metrics like response time, throughput, and error rates. 2. Analyzed deployment logs and usage patterns to identify potential optimization opportunities. 3. Fine-tuned resource allocation, adjusting instance types and scaling parameters for optimal efficiency. 4. Implemented caching mechanisms to reduce redundant computations and enhance prediction speed. 5. Employed load testing to simulate varying user loads and evaluate system behavior under different conditions. 6. Leveraged cloud platform features to enable auto-scaling, ensuring smooth performance during peak usage. 7. Reviewed and updated model dependencies regularly to ensure compatibility and security. 8. Collaborated with DevOps teams to continuously refine deployment strategies and implement best practices.	Address any issues, and enhance	8
24-05-2023	Wednesday	Model Monitoring	1. Implemented monitoring tools to continuously track deployed model's performance and health. 2. Configured alerts for critical metrics like response time, error rates, and resource utilization. 3. Integrated logging mechanisms to capture detailed information about incoming requests and responses. 4. Established regular reporting and visualization of monitoring data to enable timely decision-making and proactive issue resolution.	Monitor model performance	2
		Model Updates	1. Collected and analyzed monitoring data to gain insights into the deployed system's behavior and performance. 2. Identified trends, anomalies, or areas requiring improvement within the monitoring data. 3. Formulated actionable recommendations for updates or adjustments to optimize system efficiency, reliability, or security based	Identify areas for enhancement	3

			on the analysis. 4. Ensured that the suggested updates align with the observed issues and contribute to overall system enhancement. 5. Collaborated with relevant teams to discuss and prioritize the proposed updates for implementation.		
25-05-2023	Thursday	Performance Optimization	1. Reviewed monitoring data and user feedback to identify areas needing improvement. 2. Collaborated with the development team to plan and prioritize enhancements. 3. Implemented code changes to address issues and introduce desired improvements. 5. Conducted thorough testing to ensure the implemented changes didn't introduce new bugs. 6. Refactored existing code or added new features to align with enhancement goals. 7. Optimized algorithms, data structures, or processes to enhance system performance. 8. Integrated user interface enhancements or user experience improvements as needed. 9. Validated enhancements with real-world data and user scenarios to ensure effectiveness. 10. Documented the implemented changes and updates for reference and future maintenance.	Optimize model performance	8
		Evaluation	1. Assessed the model's performance post optimization to gauge the effectiveness of implemented changes. 2. Utilized relevant evaluation metrics, such as accuracy, precision, recall, or F1-score, to measure improvements. 3. Compared post-optimization results with pre-optimization benchmarks to quantify the impact of the enhancements.	Document optimization and results	2
26-05-2023	Friday	Task Review and Planning	1. Conducted a comprehensive review of completed tasks during Week 4 to assess progress. 2. Developed a task plan for Week 5, outlining goals, priorities, and deadlines to guide efficient work allocation and continued project advancement.	Prepare for next week's activities	1

		Communication and Updates	- Scheduled a team meeting to provide updates on project status and recent developments. - Shared information about accomplished milestones, challenges faced, and forthcoming tasks. - Ensured clear communication and alignment among team members to facilitate effective collaboration.	Discuss findings, share insights	2
		Model Performance Monitoring	- Continuously collected and analyzed real-world usage data and feedback from users. - Tracked key performance metrics such as response times, prediction accuracy, and user satisfaction. - Identified any deviations from expected behavior or performance through ongoing monitoring. - Used insights gained from monitoring to make timely adjustments, updates, or optimizations to maintain or improve the model's real-world performance.	Analyze the model's behaviour and trends	5

Week 5

Date	Day	Task	Task Description	Work Done	Hours
29-05-2023	Monday	Demo Preparation	- Ensured that all implemented enhancements and optimizations are thoroughly tested and functional. - Organized the necessary data, visuals, and documentation to showcase the system's capabilities. - Practiced the demonstration to ensure smooth execution and clarity in presenting the system's features. - Prepared explanations for each aspect of the system, highlighting its benefits, improvements, and user value to effectively convey the project's accomplishments.	Rehearse demo, prepare materials	2
		Finalization and Wrap-Up	- Conducted a comprehensive review of the entire project, examining all aspects from initial goals to final outcomes. - Gathered relevant documentation, reports, and materials to support the presentation. - Crafted a structured presentation outline, ensuring key points are covered and the flow is coherent. - Rehearsed the presentation to refine delivery, pacing, and addressing potential questions. - Anticipated possible challenges,	Summarize work, prepare for demo	4

			prepared backup plans, and ensured all technical setups were in place for a seamless presentation.		
		Project Demo	1. Showcased the project outcomes and achievements to stakeholders, presenting a summary of the journey from initial goals to the final results. 2. Highlighted key accomplishments, improvements, and the overall impact of the project on the intended goals or problem-solving objectives.	Showcase work to the team	1
		Demo Reflection and Updates	1. Reviewed feedback received from the demo presentation, considering both positive responses and areas for improvement. 2. Identified specific points of feedback related to clarity, functionality, or user experience. 3. Utilized the feedback to refine the project further, making necessary adjustments to enhance its overall quality and effectiveness.	Gather feedback, update materials	2
30-05-2023	Tuesday	Documentation Update	1. Compiled all project-related documents, including code documentation, design decisions, and implementation details. 2. Ensured that the documentation is comprehensive, organized, and ready for archival and future reference.	Review and update project report	1
		Notion Update	1. Opened the project page on Notion to access the existing information. 2. Edited the project description, objectives, and scope to reflect the latest developments. 3. Updated task lists and milestones, marking completed tasks and adding any new ones. 4. Added relevant links, attachments, or references to external resources, ensuring all necessary information is easily accessible. 5. Included the latest data visualizations, diagrams, or images to provide an up-to-date visual representation of the project. 6. Documented the outcomes of the final presentation and any feedback received from stakeholders. 7. Recorded any future plans, if applicable, indicating next steps or potential avenues for further	Document progress and achievements	8

			development. 8. Ensured that all changes are saved and organized logically within the Notion workspace.		
31-05-2023	Wednesday	Task Review and Planning	1. Reflected on the internship experience, considering accomplishments, challenges, and lessons learned. 2. Defined future career or educational goals based on the insights gained during the internship.	Reflect on experience, plan for future	2
		Reflection and Learning	1. Evaluated the skills and knowledge gained during the internship in relation to initial expectations. 2. Assessed personal growth, highlighting areas of improvement and newfound strengths. Reflected on project contributions, challenges faced, and solutions developed. 3. Considered how the internship experience aligns with long-term career aspirations and learning goals.	Summarize lessons learned, growth	2
		Communication and Updates	1. Scheduled the last team meeting to provide a final update on project outcomes and achievements. 2. Shared insights on completed tasks, accomplishments, and lessons learned throughout the project. 3. Discussed future plans, ongoing responsibilities, and opportunities for continued collaboration.	Share insights.	2

Week 5

Date	Day	Task	Task Description	Work Done	Hours
01-06-2023	Thursday	Kick-off Meeting and Goal Setting	1. Conducted a team meeting to establish clear project goals, objectives, and deliverables. 2. Collaborated with team members to ensure a shared understanding and alignment on the project's direction and purpose.	Understand project objectives	1

		Literature Review	1. Conducted online research to gather information about integrating Robyn (a tool for generating reports from Jupyter notebooks) into Python. 2. Explored available documentation, tutorials, and community forums to understand the integration process. 3. Investigated compatibility and dependencies required for seamless Robyn integration with Python programming. 4. Performed a similar research process for integrating Robyn with R, exploring relevant resources and guidelines. 5. Made notes of key findings, potential challenges, and best practices for integrating Robyn in both Python and R environments. 6. Prepared a summary of the research findings to share with the team and support informed decision-making.	Gain insights into the integration	4
		Software Installation and Setup	1. Installed the latest Python interpreter and required libraries for the project's Python development environment. 2. Set up the R environment by installing the R programming language and relevant packages for analysis and integration purposes.	Ensure consistent setup	1
		Data Gathering	1. Gathered sample data representative of the project's scope and requirements. 2. Ensured the collected data covers a range of scenarios to comprehensively test the Python and R integration.	Gather relevant datasets	1
		Environment Configuration	1. Set up the development environment by configuring the necessary IDEs (Integrated Development Environments) for both Python and R. 2. Installed any additional extensions, plugins, or packages required to streamline the integration process.	Set up IDEs, version control	1
02-06-2023	Friday	Initial Python Code Integration	1. Familiarized myself with Robyn's documentation and functionalities specific to Python integration. 2. Imported the required Robyn library and initialized the necessary components within the Python environment. 3. Created a sample Jupyter notebook to experiment with Robyn's features and generate preliminary reports. 4. Integrated sample data into the notebook to simulate real-world scenarios for report generation. 5. Customized the report templates and layouts using Robyn's Python functions to suit the project's needs. 6. Iteratively tested and refined the integration, addressing any issues or errors that arose during the process.	Implement initial Python functionality	4

		Initial R Code Integration	1. Explored Robyn's documentation to understand the steps required for integrating it into R. 2. Installed the necessary R packages and dependencies to facilitate Robyn's integration. 3. Created an R script or notebook to initiate the integration process and generate preliminary reports using Robyn. 4. Imported sample data into the R environment to simulate real-world scenarios for report generation. 5. Utilized Robyn's R functions to customize report templates and layouts according to the project's requirements. 6. Conducted iterative testing and refinement to ensure successful integration and address any encountered challenges or errors.	Implement initial R functionality	4
--	--	----------------------------	--	-----------------------------------	---

Week 6

Date	Day	Task	Task Description	Work Done	Hours
------	-----	------	------------------	-----------	-------

05-06-2023	Monday	Python-R Integration Refinement	1. Collaborated with team members to discuss the progress and initial outcomes of the Python-R integration. Identified potential areas for improvement or enhancement based on the integration's current state. 2. Analyzed the generated reports to ensure they meet the expected quality and match the project's objectives. 3. Addressed any discrepancies or inconsistencies between the Python and R integration outputs. 4. Conducted benchmarking or performance comparisons to evaluate the efficiency of the integrated solutions. 5. Fine-tuned parameters, configurations, or code snippets to optimize the integration's overall performance. 6. Ensured compatibility with different data types, formats, and use cases during the integration refinement process. 7. Conducted thorough testing on various scenarios and edge cases to validate the robustness of the integration. 8. Documented any refinements made to the integration process, including code changes, adjustments, and rationale. 9. Prepared a summary report detailing the refined Python-R integration, its benefits, and any lessons learned during the process.	Improve compatibility, structure	8
06-06-2023	Tuesday	Initial Testing	1. Executed the Python code snippets related to Robyn integration in a controlled environment. 2. Verified that the code runs without errors and generates the intended outputs. 3. Analyzed the preliminary results and reports to ensure they align with expectations. 4. Captured any issues or anomalies encountered during the initial testing for further investigation and improvement.	Verify basic functionality	2
		Debugging	1. Reviewed the outcomes of the initial testing, paying attention to any errors, discrepancies, or unexpected behaviors. 2. Analyzed code logic, configurations, and data inputs to pinpoint the root causes of the identified issues. 3. Implemented appropriate fixes or	Address bugs and glitches	4

			adjustments to the Python code to address the initial problems. Conducted subsequent testing to verify that the applied changes resolved the issues effectively.		
07-06-2023, 08-06-2023	Wednesday, Thursday	Initial R Code Integration	1. Extended the Python-Robyn integration by incorporating more complex data scenarios and use cases. 2. Leveraged Robyn's advanced features to generate detailed and informative reports from the integrated Python environment. 3. Developed Python functions or scripts that seamlessly interact with Robyn's capabilities, enhancing the integration's versatility. 4. Explored customization options within Robyn to tailor reports to specific stakeholder needs and requirements. 5. Integrated real-world data into the Python environment to evaluate the integration's effectiveness and practicality. 6. Conducted thorough testing to ensure that the expanded integration maintains stability and accuracy. Collaborated with team members to share progress, gather feedback, and ensure alignment with project goals.	Enhance Robyn (R) functionality	6
		Integration Refinement	1. Reviewed the results of testing the integrated code, including any identified errors or areas for improvement. 2. Analyzed testing feedback to understand the root causes of issues and challenges encountered. 3. Addressed bugs, inconsistencies, or performance bottlenecks by making necessary code adjustments. 4. Refined the integration logic to enhance efficiency, reliability, and overall functionality. 5. Improved error handling mechanisms to ensure the integrated code gracefully handles unexpected scenarios. 6. Conducted regression testing to verify that the code refinements didn't introduce new issues. 7. Tested the code across a range of datasets, scenarios, and user inputs to ensure robustness. 8. Collaborated with colleagues or team members to validate the effectiveness of the code refinements.	Improve integration	8

			9. Documented the refinements made, explaining the changes, their motivations, and their impact on the integration. 10. Prepared a summary report outlining the refined integrated code's benefits, improvements, and readiness for deployment.		
08-06-2023, 09-06-2023	Thursday, Friday	Unit Testing	1. Created unit test cases for specific functions or modules within the integrated Python code. 2. Designed test scenarios that cover various input combinations and edge cases to ensure comprehensive coverage. 3. Executed the unit tests to validate the correctness and reliability of the Python code's individual components. 4. Applied similar unit testing procedures to the integrated R code, verifying its functionalities and outputs. Documented the test results, highlighting successful tests and any identified issues for further resolution.	Ensure individual functionality	4
		Performance Analysis	1. Reviewed the outputs and reports generated by the integrated Python-Robyn solution. 2. Analyzed performance metrics such as processing time, memory usage, and resource consumption. 3. Identified any performance bottlenecks, inefficiencies, or areas where optimization is needed. 4. Examined the scalability of the integrated solution to determine how well it handles larger datasets or increased user demands. 5. Collected insights from the initial performance analysis to guide subsequent optimization efforts and refinements.	Identify potential bottlenecks	4
		Integration Debugging	1. Identified and diagnosed integration-related issues within the project's systems, pinpointing challenges that hindered seamless communication between different components. 2. Collaborated with relevant team members to troubleshoot and resolve	Resolve integration challenges	2

			integration problems, ensuring a smooth flow of data and functionalities. 3. Documented the encountered issues, their root causes, and the implemented solutions, creating a reference for future problem-solving and promoting continuous improvement in the integration process.		
		Task Review and Planning	1. Conducted a thorough review of the tasks accomplished in Weeks 1 and 2, assessing progress and outcomes to identify areas for improvement or adjustments. 2. Collaboratively planned tasks for Week 3, considering feedback, project priorities, and remaining goals, and allocated responsibilities among team members to ensure effective execution.	Prepare for next week's activities	1
		Communication and Updates	1. Organized a team meeting to provide updates on the project's progress, sharing accomplishments, challenges, and any roadblocks encountered. 2. Collaborated with team members to ensure a comprehensive understanding of the project's current status, fostering open discussions and addressing any issues to keep the project on track.	Discuss findings, share insights	1

Week 7

Date	Day	Task	Task Description	Work Done	Hours
12-06-2023	Monday	Integration Testing	1. Carried out thorough integration tests across different components of the project, simulating real-world scenarios to identify potential issues and ensure seamless interactions. 2. Collaborated with developers and testers to design test cases that cover various integration points, validating data flow, communication protocols, and system behavior. 3. Documented the integration test results, including any identified bugs or inconsistencies, providing a clear overview of the system's readiness for deployment and helping to guide further improvements.	Test end-to-end functionality	2

		Robyn Functionality Testing	1. Conducted comprehensive functionality tests on the Robyn system, scrutinizing its features and capabilities to ensure they aligned with the project's requirements. 2. Collaborated with the testing team to design test scenarios that covered a wide range of functionalities within the Robyn system. 3. Documented the results of the functionality tests, outlining any observed discrepancies, errors, or deviations from expected behavior, contributing to the overall assessment of the system's quality and readiness.	Verify Robyn integration	2
		Python-R Interaction Testing	1. Tested interactions between Python and R components, verifying data exchange, compatibility, and seamless communication between the two programming languages. 2. Created test cases that encompassed scenarios involving data transfer, function calls, and code execution between Python and R scripts. 3. Documented the test outcomes, highlighting successful interactions and addressing any issues or limitations encountered during the testing process.	Validate seamless communication	2
13-06-2023	Tuesday	Bug Fixes and Improvements	1. Identified and documented issues reported during testing, including bugs, glitches, and performance concerns. 2. Collaborated with relevant team members, such as developers and testers, to analyze the root causes of the identified issues and devise effective solutions. 3. Implemented fixes and improvements to address the issues, following a systematic approach to ensure that the solutions were robust and aligned with the project's goals. 4. Conducted thorough testing of the implemented solutions to verify that the issues were resolved and that no new problems arose as a result. 5. Collaborated closely with stakeholders to communicate the progress of issue resolution, manage expectations, and ensure alignment on the steps taken to address the identified problems.	Enhance code quality and stability	8
14-06-2023	Wednesday	Code Review	1. Conducted a comprehensive code review of the integrated components, examining the codebase for adherence to coding standards, best practices, and overall quality. 2. Collaborated with fellow developers to identify any potential issues such as code duplication, suboptimal algorithms, and lack of comments or documentation. 3. Provided constructive feedback and	Identify areas for further refining	2

			suggestions for improvements to enhance the code quality, readability, and maintainability, ensuring a robust and well-structured codebase.		
		Performance Optimization	1. Analyzed the integrated code to identify areas where optimization could enhance performance and efficiency, such as reducing computational complexity or minimizing resource usage. 2. Collaborated with the development team to implement optimization techniques, which could include algorithmic improvements, memory management strategies, or parallel processing approaches. 3. Conducted rigorous testing after implementing optimizations to ensure that the changes did not introduce new issues and that the overall performance of the integrated code was improved as intended.	Enhance overall performance	4
		Integration Debugging	1. Reviewed the integrated system to identify any lingering integration-related issues or bugs that might have been missed during previous testing phases. 2. Collaborated with relevant team members to diagnose and debug the identified issues, using logs, error messages, and debugging tools to pinpoint the root causes. 3. Implemented effective solutions to resolve the integration issues, ensuring that the system was functioning smoothly and that all components were interacting as intended.	Ensure seamless Python-R integration	2
15-06-2023	Thursday	Testing Evaluation	1. Conducted a comprehensive evaluation of the integration process and its outcomes, comparing the project's initial goals with the achieved results. 2. Collaborated with team to gather feedback and insights on the integrated system's functionality, performance, and usability. 3. Analyzed key performance indicators, user experience metrics, and any post-integration issues to assess the success of the integration and identify areas for further refinement or improvement.	Analyse testing outcomes	2

		Code Refinement	1. Reviewed the results of testing efforts to identify areas where code refinement was necessary to address issues or discrepancies uncovered during testing. 2. Collaborated with the development team to implement code refinements, which could include bug fixes, improvements in error handling, and adjustments to ensure alignment with desired outcomes. 3. Conducted rigorous retesting after making code refinements to verify that the identified issues were successfully resolved and that the system's functionality remained intact. 4. Documented the refinements made to the code, including the reasons behind each change, to ensure transparency and maintain a record of the code evolution for future reference.	Improve code structure and quality	6
--	--	-----------------	--	------------------------------------	---

Week 8

Date	Day	Task	Task Description	Work Done	Hours
19-06-2023	Monday	Integration Refinement	1. Analyzed the results of testing efforts to identify specific areas within the integrated code that required refinement to address reported issues or improve overall quality. 2. Collaborated with the development team to implement necessary code refinements, which might include debugging, error handling improvements, and code optimization. 3. Conducted rigorous retesting of the refined code to ensure that reported issues were successfully resolved and that the code changes did not introduce new problems. 4. Engaged in discussions with testing and development team members to ensure a clear understanding of the refined code changes, promoting effective collaboration and alignment.	Enhance integration	4
		Final Testing and Evaluation	1. Conducted a final round of comprehensive testing to assess the integrated system's functionality, performance, and reliability before its deployment. 2. Collaborated with testing and quality assurance teams to ensure that all identified issues were resolved and that the system met the predefined quality standards and user requirements.	Ensure stability and accuracy	2

20-06-2023	Tuesday	Bug Fixes and Improvements	1. Reviewed the final testing outcomes to identify any remaining issues or anomalies that might impact the integrated system's performance or functionality. 2. Collaborated with the development team to prioritize and address the identified issues, ensuring that critical problems were resolved before deployment. 3. Implemented effective solutions to resolve the issues, conducting thorough testing after each fix to verify that the problem was adequately addressed. 4. Engaged in discussions with stakeholders to communicate the progress of issue resolution, manage expectations, and ensure alignment on the actions taken. 5. Conducted final validation testing to ensure that all identified issues were successfully addressed, verifying that the integrated system was ready for deployment.	Enhance code quality and stability	8
21-06-2023	Wednesday	Final Python-R Interaction Test	1. Conducted a comprehensive interaction test to ensure that all components of the integrated system interacted seamlessly and as intended. 2. Designed test scenarios that covered a wide range of user interactions and system functionalities, identifying any potential issues related to data flow, communication, or integration. 3. Collaborated with testing and development teams to analyze the results of the interaction test, addressing any identified issues and ensuring that the integrated system provided a smooth user experience.	Validate Python-R integration	2
		Performance Analysis	1. Conducted a performance analysis after implementing code refinements and optimizations, measuring the impact of the changes on system speed, efficiency, and resource usage. 2. Collaborated with performance analysts to interpret the data and assess whether the implemented refinements had resulted in the desired improvements and met the project's performance goals.	Evaluate optimization success	2
		Deployment Preparation	1. Compiled and organized the refined codebase and required resources to ensure a smooth deployment process. 2. Collaborated with the deployment team to package the code, dependencies, and any necessary configuration files in a way that facilitates efficient and error-free deployment. 3. Conducted thorough testing on the deployment package to verify its	Package code and dependencies	2

			functionality and compatibility in the target environment, addressing any issues that might arise during deployment.		
		Deployment to Test Environment	1. Deployed the prepared code and resources to a test environment, simulating the production environment to ensure that the system functions as intended in a controlled setting. 2. Collaborated with the deployment team to execute the deployment process, closely monitoring the deployment to identify and address any issues or discrepancies that might arise during this stage.	Ensure successful deployment	2
22-06-2023	Thursday	Testing and Validation	1. Collaborated with testing and quality assurance teams to verify that the deployed system produced the expected results and behaved consistently with the requirements and specifications. 2. Analyzed the test results and compared them to the expected outcomes, addressing any discrepancies and making necessary adjustments to ensure a successful deployment.	Confirm expected behaviour	2
		Bug Fixes and Enhancements	1. Identified and documented any deployment-related issues that arose during the testing phase in the test environment. 2. Collaborated with the deployment and development teams to diagnose the root causes of the deployment issues and implement effective solutions. 3. Conducted thorough retesting after implementing the solutions to ensure that the deployment-related issues were successfully resolved and that the system was functioning as expected in the test environment.	Ensure smooth operation	6
		Accuracy Review	1. Collaborated with data scientists and domain experts to interpret the model's accuracy in the context of the project's objectives and requirements. 2. Analyzed potential factors influencing the model's accuracy, such as data quality, feature selection, functionality, performance, and compatibility.	Ensure completeness and accuracy	2
23-06-2023	Friday	Task Review and Planning	1. Engaged in a reflective process to assess the project's overall progress, achievements, and challenges, gathering insights to inform future planning and decision-making.	Prepare for the final stages of work	1

		Finalization Reflection	1. Conducted a comprehensive reflection on the overall project, evaluating its goals, outcomes, and lessons learned throughout its lifecycle. 2. Collaborated with team members to share individual perspectives and insights, fostering a collective understanding of the project's successes, areas for improvement, and potential future directions.	Review achievements and challenges	2
		Deployment to Production	1. Deployed the thoroughly tested and validated code to the production environment, ensuring that the system was ready for use by end-users. 2. Collaborated closely with the deployment team to execute the deployment process smoothly, monitoring the deployment to address any potential issues promptly. 3. Conducted post-deployment testing to verify that the system was functioning as expected in the production environment and that no new issues had arisen during deployment.	Ensure successful production deploy	4
		Real-Time Testing	1. Conducted tests to evaluate the real-time functionality of the deployed system, ensuring that it could handle and process data in real-time scenarios. 2. Collaborated with the testing team to design and execute test scenarios that simulated real-time data inputs and interactions, assessing the system's responsiveness and accuracy. 3. Analyzed the test results to verify that the real-time functionality met the project's requirements and performed as intended, addressing any discrepancies or performance issues as needed.	Interact with deployed code	4
		Communication and Updates	1. Engaged in team meeting to provide updates on the project's current status, sharing accomplishments, challenges, and milestones reached since the last update	Discuss findings, share insights	1

Week 9

Date	Day	Task	Task Description	Work Done	Hours
26-06-2023, 27-06-2023	Monday, Tuesday	Docker Performance Analysis	1. Collaborated with Docker experts and system administrators to diagnose the root causes of the performance issues, using tools like monitoring and profiling to gather relevant data. 2. Proposed and implemented optimizations, adjustments to container configurations, and	Identify bottlenecks and slowdowns	4

			resource allocation changes to mitigate the identified performance issues and enhance overall system efficiency.		
		Performance Optimization	1. Analyzed the existing Docker container setup to identify areas where efficiency improvements could be made, such as optimizing resource allocation, reducing image size, and streamlining dependencies. 2. Collaborated with Docker specialists and developers to implement optimization techniques, including optimizing container startup times, minimizing unnecessary layers, and optimizing container communication. 3. Conducted performance tests before and after optimization to measure the impact of the changes on resource usage, response times, and overall system efficiency.	Enhance container performance	8
		Troubleshooting Docker Issues	1. Identified Docker-related problems, such as container crashes, network issues, or resource allocation conflicts, through monitoring, error logs, and user feedback. 2. Collaborated with Docker experts and system administrators to diagnose the root causes of the identified problems, employing debugging tools and analysis to pinpoint issues. 3. Implemented effective solutions to resolve the Docker problems, which might include adjusting container configurations, optimizing resource allocation, or updating Docker images. 4. Conducted thorough testing after implementing solutions to validate that the identified Docker problems were successfully resolved and that the system was functioning smoothly.	Address Docker-related problems	6

28-06-2023	Wednesday	Python-R Integration Debugging	1. Investigated code breaks by analyzing error messages, stack traces, and log files to identify the root causes of the issues. Collaborated with development team members to diagnose the code breaks, discussing potential solutions and implementing fixes to restore functionality. 2. Conducted thorough testing after implementing the fixes to ensure that the code breaks were successfully addressed and that the system was functioning correctly without any regression issues.	Diagnose and repair integration issues	4
		Code Refinement	1. Reviewed the existing Python-R integration code to identify areas where improvements or optimizations could enhance the interaction between the two languages. 2. Collaborated with developers proficient in both Python and R to refine the integration code, ensuring smoother communication, efficient data exchange, and seamless execution of combined functionality.	Enhance integration stability	4
29-06-2023, 30-06-2023	Thursday, Friday	Docker Testing and Validation	1. Tested the Docker container after implementing changes or optimizations, verifying that the modifications did not introduce new issues and that the container still functioned as expected.	Confirm improved performance	1
		Integration Testing	1. Developed a comprehensive testing plan that covers various aspects of the Python-R integration. 2. Executed end-to-end tests that involve running integrated code on different datasets and scenarios.	Verify correct functionality	2
		Documentation Update	1. Added detailed information about the Python-R integration process to the project documentation. 2. Included steps, code snippets, and explanations that guide future users through the integration setup and usage.	Reflect changes in documentation	2
		Code Review and Refactor	1. Identified areas where the code structure could be improved for better modularity and maintainability. 2. Restructured code modules, functions, and classes to enhance clarity and promote efficient collaboration.	Enhance code quality and readability	1

		Task Review and Planning	1. Reviewed the tasks accomplished throughout the project, considering the initial goals and objectives. 2. Assessed the progress made in terms of completed milestones, challenges faced, and lessons learned. 3. Reflected on personal growth, skill development, and areas where improvements can be made.	Plan for the final stages of work	1
		Final Testing and Validation	1. Developed a comprehensive testing plan that outlines different testing scenarios, data inputs, and expected outcomes. 2. Executed various types of testing, including unit testing, integration testing, and end-to-end testing.	Ensure robust integration and stability	2

Week 10

Date	Day	Task	Task Description	Work Done	Hours
03-07-2023	Monday	Docker Finalization	1. Analyzed the existing Docker container setup to identify potential areas for optimization. 2. Streamlined the container configuration, including minimizing image size and reducing unnecessary dependencies. 3. Implemented best practices for Dockerfile construction and container resource management to enhance efficiency.	Ensure optimal performance	4
		Integration Finalization	1. Ensured that the Python-R integration code is fully functional and meets the project's requirements. 2. Verified that the integrated solution generates accurate reports and outputs as expected. 3. Conducted a final round of testing to confirm the stability and reliability of the integration. 4. Addressed any last-minute issues or refinements based on the final testing results.	Ensure stable and reliable integration	4

04-07-2023	Tuesday	Bug Fixes and Enhancements	1. Reviewed any outstanding issues, bugs, or challenges that were identified during testing. 2. Analyzed the root causes of these issues and their potential impact on the integrated solution. 3. Prioritized the resolution of critical issues and implemented necessary code adjustments or fixes. 4. Conducted thorough testing after implementing fixes to ensure the issues were effectively resolved.	Improve code quality and stability	8
05-07-2023	Wednesday	Code Review and Refactor	1. Examined code organization, readability, adherence to coding standards, and modularity. 2. Identified any potential code smells, redundancy, or opportunities for code optimization. 3. Provided constructive feedback and suggestions for improvement to enhance the overall code quality.	Optimize code for better maintenance	2
		Final Docker Testing	1. Executed a variety of tests to assess the container's functionality, scalability, and resource utilization. 2. Verified that the containerized application behaves consistently across different environments and configurations. 3. Tested the container's resilience to various scenarios, including high loads and unexpected failures.	Confirm optimized Docker performance	4
		Integration Testing	1. Defined a set of integration test cases that cover the interactions between different components or modules of the integrated solution. 2. Executed the integration tests to validate that the components work together seamlessly and produce the expected outcomes.	Validate Python-R integration	2
06-07-2023	Thursday	Code Review and Refactor	1. Conducted a final review of the entire codebase, considering both individual components and their interactions. 2. Identified any last-minute code improvements or optimizations that can enhance performance, readability, or maintainability. 3. Addressed any lingering issues or inconsistencies that were not resolved during previous testing phases.	Enhance code quality and readability	2
		Task Review and Reflection	1. Reflected on the entire project journey, from inception to completion, considering achievements, challenges, and personal growth throughout the process.	Summarize key learnings and insights	1

		Project Handover	1. Compiled a comprehensive guide or manual for future maintainers or users to understand and manage the project. 2. Organized all project files, resources, and documentation in a structured manner for easy access.	Document key insights and resources	2
		Knowledge Sharing	1. Shared insights gained from the project with team members, focusing on key learnings, challenges, and successes. 2. Discussed best practices, strategies, and techniques that proved valuable during the project.	Present key findings and lessons	2
07-07-2023	Friday	Notion Update	1. Opened the project page on Notion to access the existing documentation. 2. Edited and updated the project details, outcomes, and any recent changes. 3. Added new sections or information related to the final project status, integrations, and testing results. 4. Ensured that all project documentation is accurate, up-to-date, and well-organized within the Notion workspace.	Reflect changes in project progress	6
		Task Review and Reflection	1. Engaged in a comprehensive reflection on the entire project, considering goals, challenges, accomplishments, and lessons learned. 2. Evaluated the project's impact, value, and alignment with initial objectives, acknowledging areas of success and areas for improvement.	Summarize achievements and challenges	2
		Demo	1. Presented a demo showcasing the successful integration and utilization of Robyn within the project, highlighting its features and benefits.	Demo given to the stakeholders	1

Week 11

Date	Day	Task	Task Description	Work Done	Hours
10-07-2023	Monday	Geolift Planning	1. Developed a comprehensive strategy for the implementation of the Geolift project. 2. Identified key milestones and tasks required for successful implementation. 3. Allocated resources and set a timeline to ensure smooth execution.	Define key steps and objectives	1

		Environment Setup	1. Installed and configured the required software and tools for Geolift development. 2. Prepared the development environment by ensuring compatibility and proper settings. 3. Created necessary directories and structures to organize the project files.	Install required tools and libraries	2
		Geolift Code Implementation	1. Began coding the foundational logic for the Geolift project based on the established plan. 2. Developed the core algorithms and functions that drove the main features of Geolift. 3. Wrote initial code for handling geospatial data manipulation and analysis.	Begin coding the core functionality	8
11-07-2023, 12-07-2023	Tuesday, Wednesday	Initial Integration	1. Explored methods for integrating Python and R within the Geolift project. 2. Set up communication channels between Python and R to ensure seamless data exchange. 3. Wrote initial code snippets to demonstrate the interaction between the two languages	Establish basic communication	2
		Geolift Coding	1. Extended the implementation of Geolift by incorporating a range of advanced features and functionalities to enhance its capabilities. 2. Refined the existing codebase through iterative development cycles and incorporated feedback received from testing and reviews 3. Skillfully addressed various edge cases and potential challenges that arose during the development process, ensuring the robustness and reliability of the geospatial algorithms. 4. Tested the extended code thoroughly to validate its performance, reliability, and accuracy, making necessary adjustments as needed 5. Collaborated closely with the development team to maintain a cohesive approach, effectively utilizing the allocated time to achieve significant progress in the Geolift project.	Progress on implementing functionality	16

13-07-2023	Thursday	Integration Refinement	- Enhanced the integration between Python and R components by implementing optimizations that significantly improved overall performance. - Streamlined data transfer mechanisms between the two languages, effectively reducing overhead and latency for smoother communication. - Rigorously tested the integration by subjecting it to various scenarios and scenarios, ensuring reliable communication and consistent data interchange between Python and R. - Identified and resolved any compatibility issues that emerged during the testing phase, guaranteeing seamless interaction between the languages. - A dedicated effort to refine Python-R integration facilitated a more cohesive and efficient workflow, ultimately contributing to the overall success of the Geolift project.	Improve communication and stability	8
14-07-2023	Friday	Task Review and Planning	- Took a moment to review completed tasks and milestones achieved. - Evaluated if the project was on track and if any adjustments to the plan were needed. - Documented any lessons learned or insights gained during the implementation process.	Plan for the next stages of work	1
		Code Review and Refactor	- Performed a comprehensive code review to identify any potential bugs, errors, or inefficiencies. - Refactored code to adhere to coding standards and improve overall readability. - Addressed any issues raised during the review process.	Enhance code quality and readability	1
		Communication and Updates	- Communicated with the project team to provide updates on the progress of Geolift implementation. - Discussed any challenges faced and potential solutions moving forward. - Aligned everyone's understanding of the current status and next steps.	Discuss findings, share insights	1

Week 12

Date	Day	Task	Task Description	Work Done	Hours
------	-----	------	------------------	-----------	-------

17-07-2023	Monday	Geolift Coding	1. Made significant progress in the implementation of new functionality within the Geolift project. 2. Worked on refining and expanding the codebase to accommodate the evolving requirements. 3. Ensured that the newly added features seamlessly integrated with the existing logic.	Progress on implementing functionality	6
		Integration Testing	1. Conducted comprehensive testing to validate the communication and compatibility between Python and R components. 2. Ensured that data transfer mechanisms were optimized and that both languages interacted smoothly. 3. Addressed any issues that emerged during testing, enhancing the overall reliability of the integration	Verify communication and compatibility	2
18-07-2023	Tuesday	Geolift Enhancements	1. Expanded the capabilities of Geolift by introducing new features to meet the project's objectives. 2. Integrated these additional functionalities in a way that complemented the existing code architecture. 3. Conducted preliminary testing to verify the correct implementation of the added features.	Expand functionality and capabilities	4
		Testing Preparation	1. Developed a plan for testing scenarios and approach to ensure thorough validation of the codebase. 2. Identified key testing points and formulated a strategy for assessing functionality and stability.	Plan testing scenarios and approach	1
		Geolift Coding	1. Continued with the incremental development of Geolift's functionality, maintaining a steady pace of progress. 2. Ensured that the ongoing code development aligned with the established goals and timelines.	Progress on implementing functionality	1
		Comprehensive Testing	1. Executed a comprehensive testing regimen to verify the functionality and stability of the recently added features. 2. Identified and addressed any bugs or inconsistencies that surfaced during the testing phase. 3. Ensured that the codebase was	Verify functionality and stability	2

			well-prepared for subsequent stages of development.		
19-07-2023	Wednesday	Create New Docker Image	1. Designed a Docker image to encapsulate the required components and configurations for parallel processing. 2. Selected appropriate base images and defined dependencies to ensure a consistent environment. 3. Integrated parallelization libraries and tools within the Docker image to facilitate the efficient execution of tasks. 4. Configured container settings to optimize resource allocation and utilization for parallel processing. 5. Ensured the Docker image was well-documented and reproducible, providing a streamlined solution for running parallel processes in the Geolift project.	Create docker image for running the parallel process of Geolift seamlessly	8
20-07-2023	Thursday	Test New Instance	1. Rigorously executed various test scenarios on the recently developed Docker instance for parallel processing. 2. Simulated real-world workloads to gauge the efficiency and scalability of the parallelization approach. 3. Collected performance metrics, such as processing time and resource utilization, to evaluate the effectiveness of the Docker setup. 4. Analyzed the test results to identify any bottlenecks, performance issues, or areas of improvement within the parallel processing environment. 5. Used the insights gained from the analysis to fine-tune the Docker configuration and optimize the parallel processing workflow for enhanced performance.	Test parallel processes in the new instance	8
21-07-2023	Friday	Testing Analysis	1. Reviewed the outcomes of the comprehensive testing and Docker instance analysis. 2. Identified areas where the performance and efficiency of Geolift could be further enhanced.	Identify areas for improvement	2

			3. Collected data-driven insights to guide optimization efforts.		
		Bug Fixes and Enhancements	1. Dedicated time to improve the overall quality and stability of the codebase. 2. Addressed issues identified during testing and analysis, including optimizing performance bottlenecks. 3. Conducted code refactoring and adjustments to align with best practices and coding standards.	Improve code quality and stability	4
		Task Review and Planning	1. Took time to reflect on the completed tasks and the overall progress of the Geolift project. 2. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. 3. Strategized for the final stages of work, considering any lessons learned and insights gained.	Plan for the final stages of work	1
		Communication and Updates	1. Engaged in a team meeting to discuss findings and share insights from the recent development and testing phases. 2. Provided updates on progress and addressed any questions or concerns raised by team members.	Discuss findings, share insights	1

Week 13

Date	Day	Task	Task Description	Work Done	Hours
24-07-2023	Monday	Geolift Refinement	1. Focused efforts on enhancing the codebase of Geolift to improve its overall quality and stability. 2. Conducted code reviews and implemented optimizations based on established best practices. 3. Fine-tuned algorithms and functionalities to ensure smoother execution and maintainability. 4. Addressed any bugs or inconsistencies detected during the refinement process.	Enhance code quality and stability	8
25-07-2023, 26-07-2023	Tuesday, Wednesday	Integration Testing	1. Executed a series of integration tests to validate the seamless interaction between Python and R components. 2. Ensured that the communication and data exchange mechanisms were functioning correctly. 3. Verified that Python-R integration was in accordance with the project's requirements and specifications.	Validate Python-R integration	4

			4. Identified and resolved any compatibility issues that arose during the testing phase.		
		Bug Fixes and Enhancements	1. Dedicated time to meticulously identify and tackle various issues within the Geolift project. 2. Collaborated with team members to prioritize and effectively address critical issues that hindered progress. 3. Conducted comprehensive debugging and troubleshooting to isolate the root causes of reported problems. 4. Implemented targeted fixes and solutions to rectify bugs, inconsistencies, and errors in the codebase. 5. Ensured that resolved issues were thoroughly tested and validated to prevent recurrence.	Improve code quality and stability	8
		Deployment Preparation	1. Organized project resources and codebase in preparation for deploying Geolift to the Vertex AI platform. 2. Ensured that all necessary dependencies and configurations were properly set up for deployment. 3. Conducted checks to confirm that the deployment environment was consistent and ready for use.	Organize resources and code for deploy	3
27-07-2023	Thursday	Final Testing	1. Executed a comprehensive series of tests to validate the end-to-end functionality of Geolift. 2. Checked for proper integration of new features and validated that existing functionalities remained intact. 3. Verified the project's behavior against predetermined use cases and requirements.	Verify end-to-end functionality	4
		Deployment to Test Environment	1. Successfully deployed the refined Geolift project to a designated test environment. 2. Ensured that the deployment process was executed without issues or errors. 3. Verified that all components were running as expected within the test environment	Ensure successful deployment	2
		Testing and Validation	1. Conducted a focused evaluation of the deployed Geolift functionality to ensure expected behaviour. 2. Verified that the project's behaviour aligned with the desired outcomes and met project requirements. 3. Addressed any discrepancies or	Confirm expected behaviour	2

			issues identified during the testing phase.		
28-07-2023	Friday	API Testing	1. Validated the functionality of API endpoints within the Geolift project. 2. Conducted tests to ensure that the APIs responded correctly and provided the intended functionalities	Validate API functionality	2
		Notion Updates	1. Recorded the changes and progress made in the project within the Notion documentation. 2. Updated relevant sections to accurately reflect the current state of the project. 3. Ensured that project-related information was easily accessible and up to date	Reflect changes in project progress	2
		Knowledge Sharing	1. Presented key insights, findings, and lessons learned from the Geolift project to the team. 2. Shared experiences related to challenges faced, solutions implemented, and best practices discovered.	Present key findings and lessons	2
		Task Review and Reflection	1. Engaged in a thorough reflection on the overall progress of the Geolift project. 2. Summarized achievements and identified challenges encountered throughout the project's lifecycle. 3. Considered the project's alignment with initial goals and assessed any deviations.	Summarize achievements and challenges	2
		Demo	1. Conducted a comprehensive demonstration of the Geolift project's features and functionalities. 2. Presented the project's capabilities to stakeholders, highlighting its benefits and value proposition.	Demo given to the stakeholders	1

Week 14

Date	Day	Task	Task Description	Work Done	Hours
------	-----	------	------------------	-----------	-------

07-31-2023	Monday	Requirement Gathering	1. Conducted meetings, team members, and relevant departments to discuss and understand their expectations and needs for the project. 2. Held these meetings to ensure alignment between the project's goals and the stakeholders' vision. 3. Clarified key deliverables, such as reports, models, or systems, to ensure everyone had a clear understanding of what the project would produce. 4. Identified data sources and required features for the Customer Lifetime Value (CLV) mode to determine the data sources necessary for the CLV model, which may include customer databases, transaction records, and behavioral data. 5. Identified the essential features and variables needed for the model, such as customer demographics, purchase history, and customer interaction data, to calculate CLV accurately.	Project requirements gathered.	8
08-01-2023	Tuesday	Python Environment Setup	1. Installed Python and necessary packages (e.g., NumPy, Pandas, Matplotlib, Scikit-Learn) for data analysis and modeling. 2. Ensured that the environment had the appropriate version and compatibility for the project. 3. Installed Jupyter Notebook and set it up to create an interactive and easily shareable environment for data analysis. 4. Configured Jupyter Notebook extensions and themes for improved productivity.	Python environment set up.	4
08-01-2023, 08-02-2023	Tuesday, Wednesday	Explore Packages	1. Researched the "lifetimes" and "pymc" Python packages to determine their applicability for building a CLV model. 2. Focused on understanding how these packages could contribute to the project's objectives. 3. Examined the official documentation for "lifetimes" and "pymc" to gain insights into their functionalities, methods, and capabilities. 4. Looked for information that could help in integrating these packages effectively. 5. Sought out code examples and	Researched ML Packages	8

			practical demonstrations of CLV modeling using "lifetimes" and "pymc." 6. Analyzed these examples to understand how the packages can be applied in real-world scenarios.		
08-02-2023	Wednesday	Finalization of Package	1. Investigated user reviews, forum discussions, and community feedback regarding experiences with "lifetimes" and "pymc." 2. Considered the overall sentiment, satisfaction, and any identified challenges.	Selected pymc	4
08-03-2023	Thursday	Data Exploration in BigQuery	1. Utilized BigQuery, a cloud-based data warehouse, to access and examine the dataset. 2. Executed SQL queries to gain insights into the dataset's structure, including the tables, columns, and relationships, as well as the types of data it contains. 3. Scrutinized the dataset for issues such as missing values, duplicate records, outliers, or inconsistent data formats. 4. Detected anomalies or discrepancies that might affect the reliability and accuracy of the dataset. 5. Recognized specific areas within the dataset that required data preprocessing, including cleaning, transformation, or feature engineering. 6. Determined the necessary data preparation steps to improve data quality and readiness for analysis or modeling.	Data in BigQuery explored.	8

08-04-2023	Friday	Feature Extraction	1. Examined the dataset to pinpoint key features and attributes that could offer valuable insights into customer behavior. 2. Focused on aspects such as purchase history, browsing patterns, interaction frequencies, and any other relevant data points. 3. Applied data transformation methods to convert raw data into a format suitable for analysis. 4. Aggregated and structured data to facilitate the extraction of meaningful customer behavior information. 5. Investigated advanced analysis techniques, including cohort analysis to group customers, sessionization to track user interactions, and user segmentation to categorize customers based on behavior. 6. Employed these techniques to gain deeper insights into customer behavior patterns and trends. 7. Worked closely with data engineers to streamline the extraction process. 8. Addressed any technical challenges that arose during data extraction, ensuring data was accessible and efficiently processed. 9. Created documentation outlining the methods used for data extraction. 10. Detailed the data sources accessed and any preprocessing steps applied. 11. Ensured transparency and reproducibility for future analysis and to share insights with other team members.	Key attributes extracted.	6
08-04-2023	Friday	Task Review and Planning	1. Took time to reflect on the completed tasks and the overall progress of the project. 2. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. 3. Strategized for the next stages of work, considering any lessons learned and insights gained.	Plan for the next stages of work	1
08-04-2023	Friday	Communication and Updates	1. Engaged in a team meeting to discuss findings and share insights from exploration. 2. Provided updates on progress and addressed any questions or concerns raised by team members.	Discuss findings, share insights	1

Week 15

Date	Day	Task	Task Description	Work Done	Hours
08-07-2023	Monday	Data Transformation	1. Access the custom_attributes table, ensuring data is available for analysis. This table should contain information related to customer transactions, such as purchase dates, transaction amounts, and customer identifiers. 2. Clean the data by addressing any missing values, duplicates, or inconsistencies to ensure data quality and accuracy. 3. Determine the most recent transaction date for each customer. This represents the "Recency" value, indicating how recently a customer made a purchase. 4. Count the number of transactions or purchases made by each customer. This represents the "Frequency" value, indicating how often a customer makes purchases. 5. Calculate the total monetary value of transactions for each customer, which represents the "Monetary" value. This shows the cumulative spending of each customer. 6. Determine the time duration between the first purchase made by each customer and the present date. This represents the "T" value, indicating the duration of the customer's relationship with the business. 7. Combine the calculated R, F, M, and T values into a single dataset, with each row representing a unique customer.	Data prepared as rfm.	8

08-08-2023	Tuesday	Create Query	1. Analyze the structure of the custom_attributes table to understand its schema, including column names and data types. Identify which columns contain information related to customer transactions, such as purchase dates and transaction amounts. 2. Write an initial SQL query that retrieves the necessary data for calculating RFM and T values. This query should include columns for customer identifiers, purchase dates, transaction amounts, and any other relevant information. 3. Extend the SQL query to calculate the RFM and T values based on the extracted data. You can use SQL functions and operations to determine Recency (R), Frequency (F), Monetary Value (M), and Time since the first purchase (T). 4. Execute the query to ensure that it accurately calculates RFM and T values. Verify the correctness of the results by comparing them to expected outcomes.	Initial Query prepared	6
08-08-2023, 08-09-2023	Tuesday, Wednesday	Query Optimization	1. Review the query execution plan and identify potential performance bottlenecks. Look for opportunities to optimize the query to reduce execution time and resource consumption. 2. If the custom_attributes table is large, consider creating appropriate indexes and partitions on the columns involved in the query. Indexes can significantly improve query performance. 3. Apply filters or WHERE clauses to limit the data retrieved from the table to only what is necessary for RFM and T calculations. Reducing the dataset size can improve query performance. 4. Minimize the use of subqueries and complex joins in the query, as they can impact performance. Opt for simpler, straightforward SQL statements. 5. Utilize caching mechanisms or materialized views when appropriate to store precomputed results, reducing the need to rerun the query frequently. 6. Execute the optimized query and compare its performance with the	Query Optimized	6

			initial query. Ensure that the query retrieves results in an efficient and timely manner.		
08-09-2023, 08-10-2023	Wednesday, Thursday	Initialise PyMC model	1. Set up your data and define the variables that will be used in the model. These variables may include observed data, prior distributions, and any latent variables. 2. Specify the prior distributions for the model parameters. This involves defining the prior beliefs about the model's parameters before observing the data. 3. Write the likelihood function that describes the relationship between the model parameters and the observed data. This function quantifies how well the model fits the data. 4. Develop model custom utility functions and transformations, write these functions to preprocess the data or handle specific aspects of the modeling process.	PyMC Distribution function code completed	8

08-10-2023	Thursday	Develop PyMC model builder	1. Use PyMC to construct the model. This involves combining the prior distributions and likelihood functions to create a probabilistic model. 2. Choose the appropriate inference method for your model. PyMC offers various sampling methods like MCMC (Markov Chain Monte Carlo) and variational inference. Specify the method that suits your modeling needs. 3. Execute the chosen inference method to estimate the posterior distribution of the model parameters. This step involves sampling from the posterior using PyMC's sampling functions. 4. After running inference, analyze the results, including posterior distributions, trace plots, and summary statistics. Diagnose the model's convergence and check for any issues in the inference process. 5. Based on the analysis and diagnostic results, refine the model as needed. Make adjustments to the prior distributions, likelihood function, or the inference method. Iteration is often necessary to improve model performance.	Initial Code for model builder completed.	8
08-11-2023	Friday	Comparison of Different models	1. Start by researching these models in academic literature and relevant resources. Understand the theory and assumptions behind each model. 2. Prepare your data, which should include purchase history, customer behavior, and transaction data. This data is essential for parameter estimation and model fitting. 3. Study the Beta-Geometric model, which is often used to model purchase patterns. Note its assumptions, including the probability of being "alive" and making repeated purchases. 4. Explore the Pareto/NBD model, which is another popular CLTV model. Understand its assumptions, such as transaction process and customer dropout. 5. Learn about the Gamma-Gamma model, which focuses on modeling customer spending behavior. Familiarize yourself with its assumptions and how it estimates	Literature Survey Completed.	6

			monetary value. 6. Investigate how to estimate the parameters of these models using your data. This typically involves maximum likelihood estimation or Bayesian methods.		
		Task Review and Planning	1. Took time to reflect on the completed tasks and the overall progress of the project. 2. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. 3. Strategized for the next stages of work, considering any lessons learned and insights gained.	Plan for the next stages of work	1
		Communication and Updates	1. Engaged in a team meeting to discuss findings and share insights from exploration. 2. Provided updates on progress and addressed any questions or concerns raised by team members.	Discuss findings, share insights	1

Week 16

Date	Day	Task	Task Description	Work Done	Hours
------	-----	------	------------------	-----------	-------

08-14-2023	Monday	Develop Code for Beta-Geo Model	1. Specify the prior distributions for the model parameters. In a Beta-Geo model, you might use Beta distributions to model prior beliefs about parameters like the probability of making repeat purchases. 2. Write the likelihood function that describes the relationship between the model parameters and the observed data. In the Beta-Geo model, the likelihood function is based on the Negative Binomial distribution. 3. Use PyMC to construct the Beta-Geo model. This involves combining the prior distributions and likelihood function to create a probabilistic model for customer purchase behavior. 4. Choose the appropriate inference method for your model. Using PyMC's built-in sampling methods called NUTS (No-U-Turn Sampler), to estimate the posterior distribution.	Scripted initial code for Beta-Geo	8
08-15-2023	Tuesday	Develop dependent function of Beta-Geo for CLV	1. Created a function that calculated the probability that a customer had "churned" or stopped making purchases. This function was often complementary to the alive probability function. 2. Developed a function that estimated the probability of a customer dropping out within a specific time frame. This function was often based on the churn probability and considered various factors like attrition rates. 3. Created a function to estimate the expected number of future purchases a customer was likely to make. This function used a combination of alive probability, churn probability, and other model-specific parameters. 4. Developed a function that estimated the number of new customers who were expected to join in a given period. This function may have considered acquisition rates, marketing efforts, or other relevant factors. 5. If the functions depended on model parameters, we ensured there was a mechanism to estimate these parameters using	Progress on implementing functionality	8

			Bayesian techniques, maximum likelihood estimation, or other appropriate methods.		
08-16-2023	Wednesday	Bug Fixes and Enhancements	1. Reviewed any outstanding issues, bugs, or challenges that were identified during testing. 2. Analyzed the root causes of these issues and their potential impact on the integrated solution. 3. Prioritized the resolution of critical issues and implemented necessary code adjustments or fixes. 4. Conducted thorough testing after implementing fixes to ensure the issues were effectively resolved.	Improve code quality and stability	6
		Locally Test Enhancement	1. Defined a set of test cases that cover the interactions between different components or modules of the clv module. 2. Executed the tests to validate that the components work together seamlessly and produce the expected outcomes after the bug fixes.	Verified bug fixes	2
08-17-2024	Thursday	Code Review and Refactor	1. Examined code organization, readability, adherence to coding standards, and modularity. 2. Identified any potential code smells, redundancy, or opportunities for code optimization. 3. Provided constructive feedback and suggestions for improvement to enhance the overall code quality.	Optimize code for better performance	2
		Integration Testing	1. Conducted comprehensive testing to validate the communication and compatibility between Beta-Geo and its functional components. 2. Ensured that chain mechanisms were optimized and that priors interacted smoothly. 3. Addressed any issues that emerged during testing, enhancing the overall reliability of the integration	Verify communication and compatibility	2
		Code Enhancements	1. Expanded the capabilities of clv by introducing new features to meet the project's objectives. 2. Integrated these additional functionalities in a way that complemented the existing code architecture.	Expand functionality and capabilities	4

			3. Conducted preliminary testing to verify the correct implementation of the added features.		
08-18-2025	Friday	Testing Preparation	1. Developed a plan for testing scenarios and approach to ensure thorough validation of the codebase. 2. Identified key testing points and formulated a strategy for assessing functionality and stability.	Plan testing scenarios and approach	1
		Beta-geo Coding	1. Continued with the incremental development of functionality, maintaining a steady pace of progress. 2. Ensured that the ongoing code development aligned with the established goals and timelines.	Progress on implementing functionality	3
		Comprehensive Testing	1. Executed a comprehensive testing regimen to verify the functionality and stability of the recently added features. 2. Identified and addressed any bugs or inconsistencies that surfaced during the testing phase. 3. Ensured that the codebase was well-prepared for subsequent stages of development.	Verify functionality and stability	2
		Code Review and Refactor	1. Performed a comprehensive code review to identify any potential bugs, errors, or inefficiencies. 2. Refactored code to adhere to coding standards and improve overall readability. 3. Addressed any issues raised during the review process.	Enhance code quality and readability	1
		Communication and Updates	1. Communicated with the project team to provide updates on the progress of eta-geo implementation. 2. Discussed any challenges faced and potential solutions moving forward. 3. Aligned everyone's understanding of the current status and next steps.	Discuss findings, share insights	1

Week 17

Date	Day	Task	Task Description	Work Done	Hours
------	-----	------	------------------	-----------	-------

08-21-2023	Monday	Develop Code for Pareto-NBD Model	1. Define the prior distributions for the model parameters. In the context of a Pareto-NBD model, one might employ Pareto distributions to represent prior beliefs about parameters like the shape and scale of the distribution. 2. Formulate the likelihood function that characterizes the relationship between the model parameters and the observed data. In the Pareto-NBD model, the likelihood function is grounded in the Negative Binomial distribution. 3. Utilize PyMC to construct the Pareto-NBD model by amalgamating the prior distributions and likelihood function, thereby creating a probabilistic model for customer purchase behavior. 4. Select the appropriate inference method for your model. You can employ PyMC's built-in sampling methods, such as NUTS (No-U-Turn Sampler), for estimating the posterior distribution.	Scripted initial code for Pareto-NBD Model	8
------------	--------	-----------------------------------	---	--	---

08-22-2023	Tuesday	Develop dependent functions of Pareto-NBD for CLV	1. Formulated functions for assessing the likelihood of a customer having "churned" or ceased making purchases, often serving as a complement to the alive probability function. 2. Created a predictive function to gauge the probability of a customer discontinuing their engagement within a specified timeframe, taking into account factors such as churn probability and attrition rates. 3. Designed a function for forecasting the anticipated number of future purchases a customer is expected to make, utilizing a blend of alive probability, churn probability, and other model-specific parameters. 4. Constructed a function for predicting the quantity of new customers anticipated to join during each period, potentially factoring in acquisition rates, marketing efforts, and other pertinent variables. 5. When these functions rely on model parameters, established a mechanism for parameter estimation employing Bayesian techniques, maximum likelihood estimation, or other suitable methodologies.	Progress on implementing functionality	8
08-23-2023	Wednesday	Bug Fixes and Enhancements	1. Assessed any unresolved matters, bugs, or obstacles that surfaced during testing. 2. Investigated the underlying reasons for these issues and evaluated their potential repercussions on the integrated solution. 3. Gave priority to resolving critical problems and carried out essential code modifications or remedies. 4. Performed comprehensive testing after the implementation of fixes to confirm the successful resolution of the issues.	Improve code quality and stability	6

		Locally Test Enhancements	1. Established a suite of test cases that encompass the interactions among various components or modules within the CLV module. 2. Conducted the tests to verify that the components harmoniously collaborate and generate the anticipated results following the bug fixes.	Verified bug fixes	2
08-24-2024	Thursday	Code Review and Refactor	1. Evaluated the code's structure, clarity, conformity to coding standards, and level of modularity. 2. Detected any possible code issues, redundancy, or prospects for code enhancement. 3. Furnished constructive feedback and recommendations for refining the overall code quality.	Optimize code for better performance	2
		Integration Testing	1. Carried out thorough testing to confirm the effective communication and compatibility between the Pareto-NBD model and its functional components. 2. Verified the optimization of chain mechanisms and the seamless interaction of priors. 3. Resolved any issues identified during testing, thereby improving the overall reliability of the integration.	Verify communication and compatibility	2
		Code Enhancements	1. Enhanced the capabilities of CLV by introducing novel features to align with the project's goals. 2. Integrated these supplementary functionalities in a manner that harmonized with the current code architecture. 3. Performed initial testing to confirm the accurate implementation of the newly added features.	Expand functionality and capabilities	4
08-25-2025	Friday	Testing Preparation	1. Developed a plan for testing scenarios and approach to ensure thorough validation of the codebase. 2. Identified key testing points and formulated a strategy for assessing functionality and stability.	Plan testing scenarios and approach	1
		Beta-NBD Coding	1. Continued with the incremental development of functionality, maintaining a steady pace of progress. 2. Ensured that the ongoing code development aligned with the established goals and timelines.	Progress on implementing functionality	3

		Comprehensive Testing	1. Carried out an exhaustive testing process to validate the functionality and robustness of the newly introduced features. 2. Recognized and resolved any bugs or discrepancies that emerged during the testing phase. 3. Secured the codebase for smooth progression into subsequent development stages.	Verify functionality and stability	2
		Code Review and Refactor	1. Conducted an extensive code review to detect possible bugs, errors, or inefficiencies. 2. Restructured the code to comply with coding standards and enhance its overall readability. 3. Rectified any concerns or problems brought to light during the review procedure.	Enhance code quality and readability	1
		Communication and Updates	1. Maintained open communication with the project team to relay updates regarding the advancement of the Pareto-NBD model implementation. 2. Engaged in discussions about encountered challenges and proposed solutions for the way ahead. 3. Ensured that everyone had a shared understanding of the present status and the forthcoming steps in the project.	Discuss findings, share insights	1

Week 18

Date	Day	Task	Task Description	Work Done	Hours
------	-----	------	------------------	-----------	-------

08-28-2023	Monday	Develop Code for Gamma-Gamma Model	1. Specify the prior distributions for the model parameters. In a Gamma-Gamma model, you might use Gamma distributions to model prior beliefs about parameters like the shape and scale of the distribution. 2. Write the likelihood function that describes the relationship between the model parameters and the observed data. In the Gamma-Gamma model, the likelihood function is based on the Gamma distribution. 3. Use PyMC to construct the Gamma-Gamma model. This involves combining the prior distributions and likelihood function to create a probabilistic model for customer purchase behavior. 4. Choose the appropriate inference method for your model, e.g., using PyMC's NUTS (No-U-Turn Sampler) for estimating the posterior distribution.	Scripted initial code for Gamma-Gamma Model	8
08-29-2023	Tuesday	Develop dependent functions of Gamma-Gamma for CLV	1. Created functions to calculate various aspects of customer behavior, such as the expected monetary value (EMV) and the probability of making repeat purchases. 2. Developed functions to estimate the monetary value of customers and their lifetime value. 3. Ensured these functions were based on the parameters of the Gamma-Gamma model. 4. Used Bayesian techniques or maximum likelihood estimation to estimate model parameters if needed.	Progress on implementing functionality	8
08-30-2023	Wednesday	Bug Fixes and Enhancements	1. Reviewed outstanding issues, bugs, or challenges identified during testing. 2. Analyzed the root causes of these issues and their impact on the integrated solution. 3. Prioritized and implemented necessary code adjustments or fixes. 4. Conducted comprehensive testing to ensure the issues were effectively resolved.	Improve code quality and stability	6

		Locally Test Enhancements	1. Defined a set of test cases to cover interactions between different components or modules of the CLV model. 2. Executed tests to validate that components work together seamlessly and produce expected outcomes after bug fixes.	Verified bug fixes	2
08-31-2023	Thursday	Code Review and Refactor	1. Examined code organization, readability, adherence to coding standards, and modularity. 2. Identified potential code smells, redundancy, or opportunities for code optimization. 3. Provided constructive feedback and suggestions for improvement to enhance overall code quality.	Optimize code for better performance	2
		Integration Testing	1. Conducted comprehensive testing to validate the communication and compatibility between the Gamma-Gamma model and its functional components. 2. Ensured that parameters and priors interacted smoothly. 3. Addressed any issues emerging during testing to enhance overall integration reliability.	Verify communication and compatibility	2
		Code Enhancements	1. Expanded the capabilities of CLV by introducing new features aligned with the project's objectives. 2. Integrated these additional functionalities in a way that complements the existing code architecture. 3. Conducted preliminary testing to verify the correct implementation of the added features.	Expand functionality and capabilities	4
09-01-2023	Friday	Testing Preparation	1. Developed a plan for testing scenarios and approach to ensure thorough validation of the codebase. 2. Identified key testing points and formulated a strategy for assessing functionality and stability.	Plan testing scenarios and approach	1
		Gamma-Gamma Coding	1. Continued with incremental development of functionality, maintaining progress aligned with established goals and timelines. 2. Ensured ongoing code	Progress on implementing functionality	3

			development aligns with project objectives.		
		Comprehensive Testing	1. Executed a comprehensive testing regimen to verify functionality and stability of the recently added features. 2. Identified and addressed any bugs or inconsistencies surfaced during testing. 3. Ensured the codebase was well-prepared for subsequent development stages.	Verify functionality and stability	2
		Code Review and Refactor	1. Performed a comprehensive code review to identify potential bugs, errors, or inefficiencies. 2. Refactored code to adhere to coding standards and improve overall readability. 3. Addressed issues raised during the review process.	Enhance code quality and readability	1
		Communication and Updates	1. Communicated with the project team to provide updates on the progress of the Gamma-Gamma model implementation. 2. Discussed challenges faced and potential solutions moving forward. 3. Aligned everyone's understanding of the current status and next steps.	Discuss findings, share insights	1

Week 19

Date	Day	Task	Task Description	Work Done	Hours
09-04-2023	Monday	Integration of Models	1. Integrate the Beta-Geo, Pareto-NBD, and Gamma-Gamma models into a unified framework. Ensure that the models work together seamlessly to provide a comprehensive customer lifetime value (CLV) prediction. 2. Develop a data pipeline to feed customer data into the integrated model.	Successfully integrated models	8
09-05-2023	Tuesday	API Development	1. Design and develop a RESTful API that exposes endpoints for predicting CLV using the integrated models. 2. Implement error handling and request validation to ensure the API is robust and user-friendly.	Fully functional CLV prediction API	6

		Add API Validations	1. Checking for required fields: Ensure that all mandatory fields are present in the input data. 2. Data type validation: Confirm that data types match the expected types (e.g., strings, numbers, dates). 3. Range and boundary checks: Validate that numerical values fall within acceptable ranges. 4. Format checks: Ensure that data adheres to specific formats (e.g., email addresses, phone numbers). 5. Cross-field validations: Check for consistency and relationships between different fields. 6. Whitelisting or blacklisting: Validate data against predefined lists of acceptable or prohibited values.	Function in place for API validation	2
09-06-2023	Wednesday	API Testing and Validation	1. Conduct thorough testing of the API to validate its functionality and reliability. 2. Create test cases to cover various use cases and edge scenarios. 3. Ensure that the API uploads accurate CLV predictions based on input data.	Verified and validated API	6
		Deployment to Vertex AI	1. Prepare the integrated model and API for deployment in Google Vertex AI. 2. Configure the necessary resources and environment for deployment. 3. Deploy the API to Vertex AI, ensuring scalability and availability.	Deployed CLV prediction API to Vertex AI	2
09-07-2023	Thursday	Testing in Vertex AI Environment	1. Conduct testing of the deployed API within the Vertex AI environment. Ensure that it can handle real-time requests and data. 2. Verify that the models perform as expected in a production setting.	Tested and verified API in Vertex AI	8
09-08-2023	Friday	Demo Preparation	1. Prepare a demo environment with sample data and use cases. 2. Create a presentation to showcase the capabilities of the integrated CLV prediction system and the deployed API.	Demo environment and presentation prepared	2

		CLV Prediction Demo	1. Conduct a live demonstration of the CLV prediction system to stakeholders and team members. 2. Showcase the API's capabilities in making accurate CLV predictions using real and simulated data.	Successful CLV prediction demo	1
		Post-Demo Adjustments and Final Checks	1. Address any issues or feedback received during the demo. 2. Perform final checks to ensure the system is running smoothly in the Vertex AI environment. 3. Prepare for potential production use.	Refined and ready for production use	2
		Documentation and Knowledge Sharing	1. Create comprehensive documentation for the integrated system, API endpoints, and model usage. 2. Share knowledge with the team to ensure they can maintain and further develop the system.	Well-documented system and shared knowledge	3

Week 20

Date	Day	Task	Task Description	Work Done	Hours
09-11-2023	Monday	Requirement Gathering	1. Meet with stakeholders to gather detailed requirements for the next purchase day prediction system. 2. Identify key features, user expectations, and any specific business rules that need to be considered.	Comprehensive list of system requirements	2
		Environment Setup	1. Set up the development environment, including necessary software, libraries, and databases. 2. Configure version control and establish collaboration tools for the development team.	Fully configured development environment	4
		BigQuery Connection	1. Connect successfully to BigQuery. 2. Explore data fields required for the exploration and EDA.	Connection established to BigQuery	2
09-12-2023	Tuesday	Data Collection and Cleaning	1. Acquire relevant historical data on customer purchase patterns. 2. Clean and preprocess the data to remove any inconsistencies or missing values.	Cleaned and prepared dataset for analysis	3
		Exploratory Data Analysis (EDA)	1. Conduct EDA to gain insights into the distribution of purchase intervals. 2. Identify patterns, trends, and outliers in the data.	In-depth understanding of the dataset	5

09-13-2023	Wednesday	Research on Models and Methods Exploration	1. Explored various machine learning models for predicting the "next purchase day." 2. Researched methods and approaches for time-series prediction. 3. Investigated relevant literature and resources to understand best practices in the field. 4. Explored potential algorithms and techniques specifically tailored for the task at hand.	Models and methods explored.	8
09-14-2023	Thursday	Choosing the Best Methods and Start Coding	1. Evaluated insights from Exploratory Data Analysis (EDA) to inform the selection of methods for predicting the "next purchase day." 2. Initiated the coding process for the initial version of the prediction model using the chosen methods. 3. Implemented basic functionality for data input, feature extraction, and model training in the coding process.	Initial coding started.	8
09-15-2023	Friday	Continue Coding for Next Purchase Day	1. Began coding the foundational logic for the functions and utils based on the established plan. 2. Developed the core algorithms and functions that drove the main features of next purchase. 3. Wrote initial code for handling customer data manipulation and analysis.	Functions and utils created in code	6
		Task Review and Planning	1. Took time to reflect on the completed tasks and the overall progress of the project. 2. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. 3. Strategized for the next stages of work, considering any lessons learned and insights gained.	Plan for the final stages of work	1
		Communication and Updates	1. Engaged in a team meeting to discuss findings and share insights from exploration. 2. Provided updates on progress and addressed any questions or concerns raised by team members.	Discuss findings, share insights	1

Week 21

Date	Day	Task	Task Description	Work Done	Hours
------	-----	------	------------------	-----------	-------

09-18-2023	Monday	Code Review and Refactor	1. Examined code organization, readability, adherence to coding standards, and modularity. 2. Identified any potential code smells, redundancy, or opportunities for code optimization. 3. Provided constructive feedback and suggestions for improvement to enhance the overall code quality.	Optimize code for better performance	2
		Integration of Function and Utils	1. Integrate the min, max, average, and monetary bucket models into a unified framework. Ensure that the models work together seamlessly to provide a comprehensive next purchase day (NPD) prediction. 2. Develop a data pipeline to feed customer data into the integrated model.	Successfully integrated functions	2
		Testing	1. Conducted initial testing on the developed model to promptly identify any issues or errors. 2. Checked for compatibility and performance concerns during the testing phase. 3. Adjusted the code as necessary based on the results and findings from the initial testing.	Initial testing completed.	4
09-19-2023	Tuesday	API Creation	1. Developed an API to make the "next purchase day" model accessible. 2. Implemented endpoints to facilitate the reception of input data and the delivery of predictions. 3. Ensured comprehensive documentation for the proper usage of the API.	API created.	4
		API Testing	1. Conducted testing on the API to ensure its proper functionality. 2. Checked for potential security vulnerabilities and addressed any identified issues. 3. Tested various input scenarios to validate the model's responses and enhance robustness.	API testing completed.	4
09-20-2023	Wednesday	Integration with Existing Prediction Container	1. Integrated the "next purchase day" model with the existing prediction container or environment. 2. Ensured seamless communication and compatibility with other components in the system. 3. Addressed any challenges related to integration that emerged during the process.	Integration completed.	6
09-20-2023, 09-21-2023	Wednesday, Thursday	Integration Testing	1. Conducted comprehensive testing on the integrated system to assess data flow, communication, and overall system performance. 2. Checked for any issues or anomalies identified during the integration testing phase. 3. Addressed and resolved issues to ensure	Integration testing completed.	4

			the smooth functioning of the integrated system.		
09-21-2023	Thursday	Optimization and Testing	1. Optimized both the code and model to enhance overall performance. 2. Conducted additional testing to verify that the optimized version maintained accuracy. 3. Addressed any remaining performance or efficiency concerns identified during testing. 4. Fine-tuned the implementation to achieve better computational efficiency. 5. Ensured that the optimized code and model continued to meet accuracy standards.	Optimization and testing done.	6
09-22-2023	Friday	Bug Fixes	1. Addressed and resolved any bugs or issues uncovered during testing. 2. Conducted thorough debugging processes to enhance the reliability of the model. 3. Implemented necessary fixes for the issues identified during the debugging phase.	Bugs fixed.	5
		Deployment on Vertex AI	1. Deployed the "next purchase day" model on Google Cloud Vertex AI. 2. Configured essential settings for a seamless deployment experience. 3. Ensured proper integration with Vertex AI services to optimize functionality and performance.	Model deployed on Vertex AI.	3

Week 22

Date	Day	Task	Task Description	Work Done	Hours
09-25-2023	Monday	Demo	1. Conducted a demonstration of the deployed "next purchase day" model. 2. Showcased key features and functionalities to stakeholders during the demonstration. 3. Gathered feedback from stakeholders and addressed any questions or concerns raised during the demonstration.	Successful demo conducted.	2

		Documentation	1. Developed comprehensive documentation for the "next purchase day" model. 2. Documented the model architecture, API endpoints, usage instructions, and troubleshooting steps. 3. Ensured the documentation was clear, concise, and accessible for ease of understanding and reference.	Documentation completed.	6
09-26-2023	Tuesday	Requirement Gathering	1. Conducted meetings with relevant stakeholders to collect requirements for the "Affinity to Buy" model. 2. Discussed expectations, project goals, and key deliverables with stakeholders during the meetings. 3. Clarified and documented the necessary features and functionalities essential for the successful implementation of the model.	Requirements gathered.	2
09-27-2023	Tuesday	Virtual Env Setup	1. Installed and configured the virtual environment for the project. 2. Set up essential Python packages, such as XGBoost and Hyperopt, to facilitate model development. 3. Ensured compatibility and version control within the virtual environment. 4. Verified the proper functioning of the configured environment for seamless model development.	Virtual environment set up.	4
		Doubt Clarification	1. Sought clarification from stakeholders to ensure a thorough understanding of gathered requirements. 2. Addressed any uncertainties or doubts regarding the project requirements through further discussions with stakeholders.	Clarified Doubts and Concerns	2

09-28-2023	Wednesday	Research on Different Models	1. Conducted a comparative analysis of various models, including neural networks, logistic regression, and XGBoost, to determine their effectiveness in predicting affinity to buy. 2. Utilized dummy data for testing and benchmarking different models to evaluate their performance under similar conditions. 3. Examined and compared the strengths and weaknesses of each model in the context of the affinity to buy prediction task. 4. Considered metrics such as accuracy, precision, recall, and F1 score to assess and contrast the models' performance on the dummy data.	Comparison completed for different models.	5
		Extraction of Customer Data from BigQuery	1. Leveraged BigQuery to extract pertinent customer data for the "Affinity to Buy" task. 2. Executed SQL queries to retrieve both features and labels required for training the model. 3. Checked and cleaned the extracted data to identify and rectify any inconsistencies or missing values, ensuring data quality for subsequent modeling.	Customer data extracted.	3

09-29-2023	Thursday	Research on XGBoost and Hyperopt	1. Explored XGBoost as a potential classifier for the "Affinity to Buy" task. 2. Conducted research on the Hyperopt library, focusing on its capabilities in hyperparameter optimization. 3. Investigated best practices for tuning XGBoost using Hyperopt to enhance model performance. 4. Gathered information on key hyperparameters relevant to optimizing XGBoost for the specific task. 5. Compiled insights on the interplay between XGBoost, Hyperopt, and hyperparameter tuning best practices for informed model development.	XGBoost and Hyperopt explored.	8
09-30-2023	Friday	EDA on the Extracted Data	1. Conducted exploratory data analysis (EDA) on the extracted customer data to gain insights into its characteristics. 2. Examined data distributions to understand the spread and central tendencies of key features. 3. Identified patterns within the data through various statistical analyses during the EDA. 4. Visualized key features to provide a comprehensive understanding of the dataset and preprocessed the data for input into the XGBoost model.	EDA on extracted data.	5
		Task Review and Planning	1. Took time to reflect on the completed tasks and the overall progress of the project. 2. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. 3. Strategized for the next stages of work, considering	Plan for the next stages of work	1

			any lessons learned and insights gained.		
		Communication and Updates	1. Engaged in a team meeting to discuss findings and share insights from exploration. 2. Provided updates on progress and addressed any questions or concerns raised by team members.	Discuss findings, share insights	1

Week 23

Date	Day	Task	Task Description	Work Done	Hours
10-02-2023	Monday	XGBoost Model Development	1. Implemented the XGBoost classifier for the "Affinity to Buy" task. 2. Tuned initial hyperparameters using default settings as a starting point for optimization. 3. Conducted training of the XGBoost model on the preprocessed data to learn patterns and relationships. 4. Monitored the training process to ensure convergence and assessed preliminary model performance. 5. Evaluated the model's performance metrics to gauge its effectiveness in	XGBoost model developed.	8

			predicting affinity to buy. 6. Iteratively refined hyperparameters and retrained the model as needed to enhance predictive accuracy.		
10-03-2023	Tuesday	Hyperparameter Optimization with Hyperopt	1. Employed Hyperopt for hyperparameter optimization of the XGBoost model. 2. Defined the search space, specifying the range of values for hyperparameters subject to optimization. 3. Conducted multiple iterations using Hyperopt to systematically explore the search space and identify the optimal hyperparameter configuration. 4. Leveraged the results of the optimization process to fine-tune the XGBoost model for improved performance on the "Affinity to Buy" task.	Hyperparameter optimization completed.	8
10-04-2023	Wednesday	Testing	1. Conducted testing on the developed XGBoost model to assess its performance. 2. Evaluated the model's performance using dedicated test data to ensure generalizability. 3. Adjusted model settings based on testing results to enhance predictive accuracy and robustness. 4. Iteratively refined the model through testing and adjustment cycles to achieve optimal performance on the "Affinity to Buy" task.	Testing completed.	3

		API Creation	1. Developed an API to facilitate accessibility to the "Affinity to Buy" model. 2. Implemented endpoints within the API to handle the reception of input data and the delivery of predictions. 3. Ensured comprehensive documentation was provided for the proper usage of the API, including input formats and expected outputs.	API created.	5
10-05-2023	Thursday	API Testing	1. Conducted testing on the API to verify its proper functionality and responsiveness. 2. Checked for potential security vulnerabilities, addressing any identified issues to enhance the API's security. 3. Tested various input scenarios to validate the XGBoost model's responses and ensure the robustness of the API.	API testing completed.	2
		Integration with Existing Prediction Container	1. Integrated the "Affinity to Buy" XGBoost model with the existing prediction container or environment. 2. Ensured seamless communication and compatibility with other components within the system. 3. Addressed any challenges related to integration that emerged during the process, ensuring a cohesive and functional integration.	Integration completed.	4
		Integration Testing	1. Conducted comprehensive testing on the integrated system to assess data flow, communication, and overall system performance. 2. Checked for any issues or irregularities identified during the integration testing phase. 3. Addressed and resolved issues to ensure the smooth functioning and reliability of the integrated system.	Integration testing completed.	2

10-06-2023	Friday	Optimization and Testing	1. Optimized the XGBoost model to enhance overall performance. 2. Conducted additional testing to validate that the optimized version-maintained accuracy. 3. Addressed any remaining performance or efficiency concerns identified during testing to ensure optimal functioning of the model.	Optimization and testing done.	2
		Bug Fixes	1. Addressed and resolved any bugs or issues identified during testing to enhance model reliability. 2. Conducted thorough debugging processes to ensure the XGBoost model's robustness. 3. Implemented necessary fixes for the issues identified during the debugging phase. 4. Ensured the final version of the model is free from critical issues and operates reliably.	Bugs fixed.	4
		Deployment on Vertex AI	1. Deployed the "Affinity to Buy" XGBoost model on Google Cloud Vertex AI. 2. Configured essential settings for a seamless deployment experience. 3. Ensured proper integration with Vertex AI services to optimize functionality and performance.	Model deployed on Vertex AI.	2

Week 24

Date	Day	Task	Task Description	Work Done	Hours
10-09-2023	Monday	Demo	1. Conducted a demonstration of the deployed XGBoost model. 2. Showcased key features and functionalities to stakeholders during the demonstration. 3. Gathered feedback from stakeholders and addressed any questions or concerns raised during the demonstration. 4. Ensured a comprehensive understanding and satisfaction among stakeholders regarding the XGBoost model's capabilities and performance.	Successful demo conducted.	2

		Documentation	1. Developed comprehensive documentation for the "Affinity to Buy" XGBoost model. 2. Documented the model architecture, API endpoints, hyperparameter settings, usage instructions, and troubleshooting steps. 3. Ensured the documentation was clear, concise, and easily accessible to facilitate understanding and reference.	Documentation completed.	6
10-10-2023	Tuesday	Requirement Gathering	1. Conducted meetings with relevant stakeholders to gather requirements for the "Propensity to Buy" model. 2. Engaged in discussions to understand expectations, project goals, and key deliverables with stakeholders during the meetings. 3. Clarified and documented the necessary features and functionalities essential for the successful implementation of the "Propensity to Buy" model.	Requirements gathered.	2
		Virtual Env Setup	1. Installed and configured the virtual environment for the project. 2. Set up essential Python packages required for model development within the virtual environment. 3. Ensured compatibility and implemented version control for the virtual environment to maintain consistency and reproducibility.	Virtual environment set up.	4
10-11-2023	Wednesday	Research on Propensity to Buy	1. Explored relevant literature and resources to gain insights into building models for "Propensity to Buy." 2. Investigated various machine learning algorithms suitable for the task, considering their strengths and weaknesses. 3. Conducted research on best practices for feature engineering to enhance the predictive capabilities of the model. 4. Explored recommended approaches for model training to optimize performance in predicting propensity to buy.	Propensity to Buy research done.	8
10-12-2023	Thursday	Extraction of Customer Data from BigQuery	1. Utilized BigQuery to extract relevant customer data for the "Propensity to Buy" task. 2. Executed SQL queries to retrieve both features and labels necessary for training the model. 3. Checked and cleaned the extracted data to identify and rectify any inconsistencies or missing values,	Customer data extracted.	4

			ensuring data quality for subsequent model training.		
		EDA on the Extracted Data	1. Conducted exploratory data analysis (EDA) on the extracted customer data to understand its characteristics. 2. Examined data distributions, identified patterns, and visualized key features during the EDA process. 3. Preprocessed the data to ensure it was appropriately formatted and ready for input into the "Propensity to Buy" model.	EDA on extracted data.	4
	Friday	Study for Best Model	1. Explored literature and resources to identify the best models for predicting propensity to buy. 2. Investigated machine learning algorithms known for their effectiveness in propensity modeling, such as logistic regression, decision trees, and ensemble methods. 3. Examined recent research papers and industry practices to stay updated on emerging models and techniques for propensity prediction. 4. Explored the use of advanced techniques like gradient boosting or deep learning for propensity modeling, considering their potential advantages. 5. Reviewed case studies and success stories to understand real-world applications and performance of different models in predicting propensity to buy. 6. Considered the scalability, interpretability, and ease of implementation of various models to determine their suitability for the specific context of propensity to buy prediction.	Best Model Selected	6
		Task Review and Planning	1. Took time to reflect on the completed tasks and the overall progress of the project. 2. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. 3. Strategized for the next stages of work, considering any lessons learned and insights gained.	Plan for the next stages of work	1
10-12-2023					

		Communication and Updates	1. Engaged in a team meeting to discuss findings and share insights from exploration. 2. Provided updates on progress and addressed any questions or concerns raised by team members.	Discuss findings, share insights	1
--	--	---------------------------	--	----------------------------------	---

Week 25

Date	Day	Task	Task Description	Work Done	Hours
10-16-2023	Monday	Model Development	1. Implemented the machine learning model for "Propensity to Buy," emphasizing feature engineering to enhance predictive capabilities. 2. Conducted thorough model training on the preprocessed data to learn patterns and relationships. 3. Utilized hyperparameter tuning to optimize the model's performance, adjusting parameters for better predictive accuracy. 4. Iteratively refined hyperparameters based on model evaluation metrics to achieve optimal results. 5. Considered cross-validation techniques to ensure the robustness of the model across different subsets of the data. 6. Evaluated the impact of various feature engineering strategies on model performance. 7. Explored different algorithms and ensemble methods to identify the most effective approach for the "Propensity to Buy" task. 8. Documented the implemented model's architecture, hyperparameter settings, and key considerations for future reference.	Propensity to Buy model developed.	8
10-17-2023	Tuesday	Continue Coding	1. Developed code for data preprocessing, handling missing values, and encoding categorical variables to prepare the data for model training. 2. Implemented feature engineering techniques, such as creating new variables or transforming existing ones, to capture relevant patterns for propensity prediction. 3. Utilized a machine learning library, such as scikit-learn, to build and train the propensity model. 4. Incorporated cross-validation strategies to assess the model's generalization performance and	Coding Completed	6

10-18-2023			mitigate overfitting. 5. Implemented a well-designed pipeline, encapsulating data preprocessing, model training, and evaluation steps, for clarity and reproducibility in future iterations.		
		Testing	1. Conducted thorough testing on the developed propensity model to assess its performance. 2. Evaluated key metrics such as accuracy, precision, recall, and other relevant indicators to gauge predictive effectiveness. 3. Adjusted model settings, including hyperparameters or feature engineering strategies, based on testing results to enhance overall performance. 4. Iteratively refined the model through testing and adjustment cycles to achieve optimal predictive accuracy for the "Propensity to Buy" task.	Testing completed.	2
	Wednesday	API Creation	1. Developed an API to facilitate accessibility to the "Propensity to Buy" model. 2. Implemented endpoints within the API to handle the reception of input data and the delivery of predictions. 3. Ensured comprehensive documentation was provided for the proper usage of the API, including input formats, and expected outputs.	API created.	2
		Integration with Existing Prediction Container	1. Integrated the "Propensity to Buy" model with the existing prediction container or environment. 2. Ensured seamless communication and compatibility with other components within the system. 3. Addressed any challenges related to integration that emerged during the process, ensuring a cohesive and functional integration.	Integration completed.	4
		Integration Testing	1. Conducted comprehensive testing on the integrated system to assess data flow, communication, and overall system performance. 2. Checked for any issues or irregularities identified during the integration testing phase. 3. Addressed and resolved issues to ensure the smooth functioning and reliability of the integrated system.	Integration testing completed.	2

09-19-2023	Thursday	Optimization and Testing	1. Optimized the "Propensity to Buy" model to enhance overall performance. 2. Conducted additional testing to validate that the optimized version-maintained accuracy. 3. Addressed any remaining performance or efficiency concerns identified during testing to ensure optimal functioning of the model.	Optimization and testing done.	4
		Bug Fixes	1. Addressed and resolved any bugs or issues identified during testing to enhance model reliability. 2. Conducted thorough debugging processes to ensure the "Propensity to Buy" model's robustness. 3. Implemented necessary fixes for the issues identified during the debugging phase. 4. Ensured the final version of the model is free from critical issues and operates reliably.	Bugs fixed.	4
09-20-2023	Friday	Deployment on Vertex AI	1. Deployed the "Propensity to Buy" model on Google Cloud Vertex AI. 2. Configured essential settings for a seamless deployment experience. 3. Ensured proper integration with Vertex AI services to optimize functionality and performance.	Model deployed on Vertex AI.	2
		Demo	1. Conducted a demonstration of the deployed "Propensity to Buy" model. 2. Showcased key features and functionalities to stakeholders during the demonstration. 3. Gathered feedback from stakeholders and addressed any questions or concerns raised during the demonstration. 4. Ensured a comprehensive understanding and satisfaction among stakeholders regarding the "Propensity to Buy" model's capabilities and performance.	Successful demo conducted.	2
		Documentation	1. Developed comprehensive documentation for the "Propensity to Buy" model. 2. Documented the model architecture, API endpoints, feature engineering strategies, usage instructions, and troubleshooting steps. 3. Ensured the documentation was clear, concise, and easily accessible for stakeholders, facilitating understanding and reference.	Documentation completed.	4